9

ME1100 Thermodynamics

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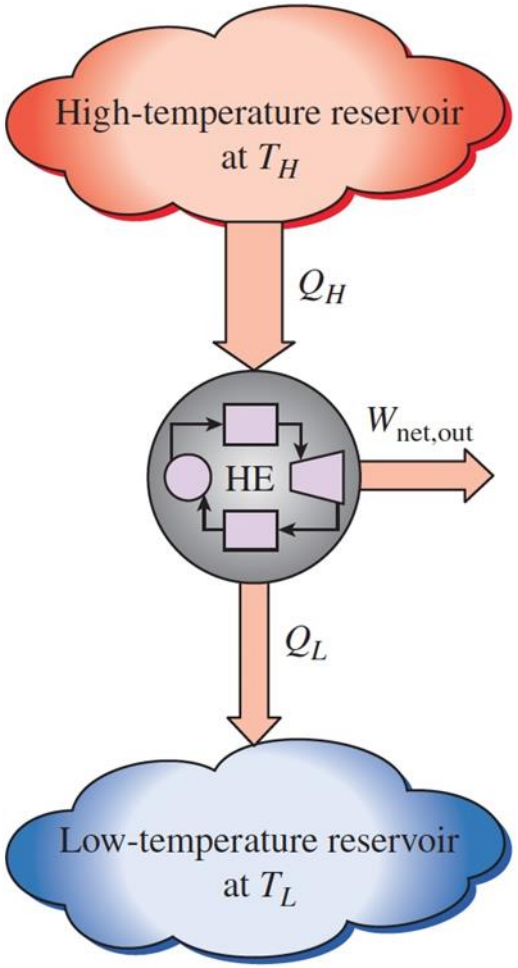
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Thermodynamics

Entropy

Entropy: From 2nd Law

Reversible heat engines



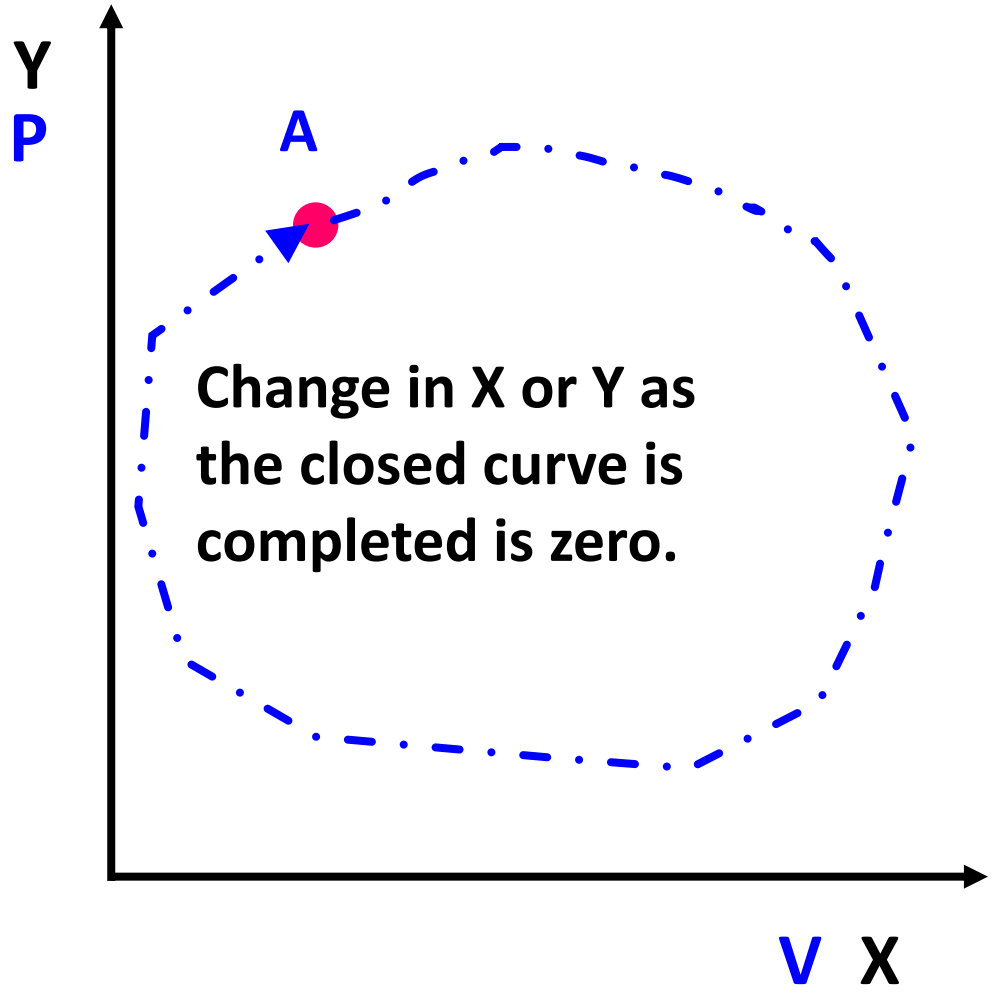
$$\frac{Q_{rev\,H}}{Q_{rev\,L}} = \frac{T_H}{T_L}$$

$$\frac{Q_{rev\,H}}{T_H} - \frac{Q_{rev\,L}}{T_L} = 0$$

$$\frac{Q_{rev\,H}}{T_H} + \left(\frac{-Q_{rev\,L}}{T_L}\right) = 0$$

$$\sum_{cycle}\left(\frac{Q}{T}\right)_{rev} = 0$$

$$\oint\left(\frac{\delta Q}{T}\right)_{rev} = 0$$



Change in thermodynamic property, which is a state (point) function, over a **thermodynamic cycle** is **zero**.

Entropy: Definition

$\frac{\delta Q_{rev}}{T}$ = **A property (rather, a change in property brought about by the transfer of heat).**

$$dS = \frac{\delta Q_{rev}}{T} \Rightarrow \Delta S = \int \frac{\delta Q_{rev}}{T} \Rightarrow S_2 - S_1 = \int_1^2 \frac{\delta Q_{rev}}{T}$$

1st Law of Thermodynamics → Energy

S: Entropy (extensive quantity), Unit: kJ/K
s: specific entropy, Unit: kJ/kg-K

2nd Law of Thermodynamics → Entropy

Unlike energy, entropy is not conserved and thus there is no such concept as the Conservation of Entropy

Since entropy is a state function even if the process between the same two states is irreversible the change in entropy is the same between these two states (points)

Entropy: Molecular picture

Entropy is associated with the degree of randomness of the system.

for three phases of a pure substance, the degree of randomness increases as it goes from solid through liquid to gas phases

Solid state: mean positions of molecules are fixed although the molecules may undergo random vibrations.

Liquid state: molecules slowly move

Vapour state: molecules move at high velocities of the order of a few hundred meters per second.

In terms of entropy, it can be said that $S_{gas} > S_{liquid} > S_{solid}$

Solid **HEAT** → **Liquid** **HEAT** → **Gas**

$$dS = \frac{\delta Q_{rev}}{T}$$

Heat added → sign is positive → dS > 0

Entropy: Molecular picture

When a gas **expands slowly** the molecules **move in a direction perpendicular** to the boundary that moves.

When a piece of **iron is magnetized** the **molecules align themselves** in a particular direction



Thus **work involves order** and when there are **no irreversibilities associated** **there is** **no increase in entropy due to work transfer**

When we **transfer heat to a substance**, the **energy transfer** due to temperature difference at the contact surface is due to the **random molecular collisions at the boundary**

This **increases** the **randomness in the system** and hence the **entropy.**

Thus, molecular chaos increases when we transfer heat to a substance which results in an increase in entropy

Summary of the last class

$$\oint \left(\frac{\delta Q}{T}\right)_{\text{rev}} = 0$$

$$dS = \frac{\delta Q_{\text{rev}}}{T} \Rightarrow \Delta S = \int \frac{\delta Q_{\text{rev}}}{T} \Rightarrow S_2 - S_1 = \int_1^2 \frac{\delta Q_{\text{rev}}}{T}$$

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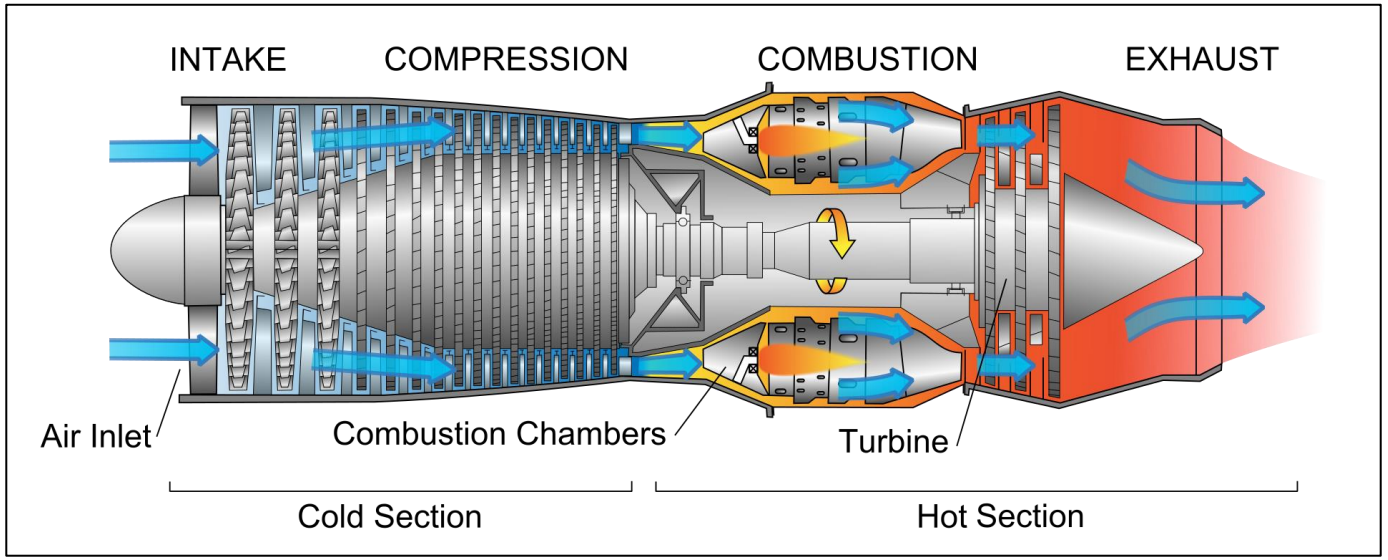
Solid HEAT → **Liquid** HEAT → **Gas**

$$dS = \frac{\delta Q_{\text{rev}}}{T}$$

Heat added → sign is positive → dS > 0

Generally, we are interested in processes

Jet engine of an aircraft



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https://commons.wikimedia.org/w/index.php?title=File:Airstrikes_in_Syria_140923-F-UL677-654.jpg&oldid=398898133.

Any reversible process

Let us consider any reversible process, i.e.

1-2: Any reversible process

a-a': Reversible adiabatic passing through 1

b-b': Reversible adiabatic passing through 2

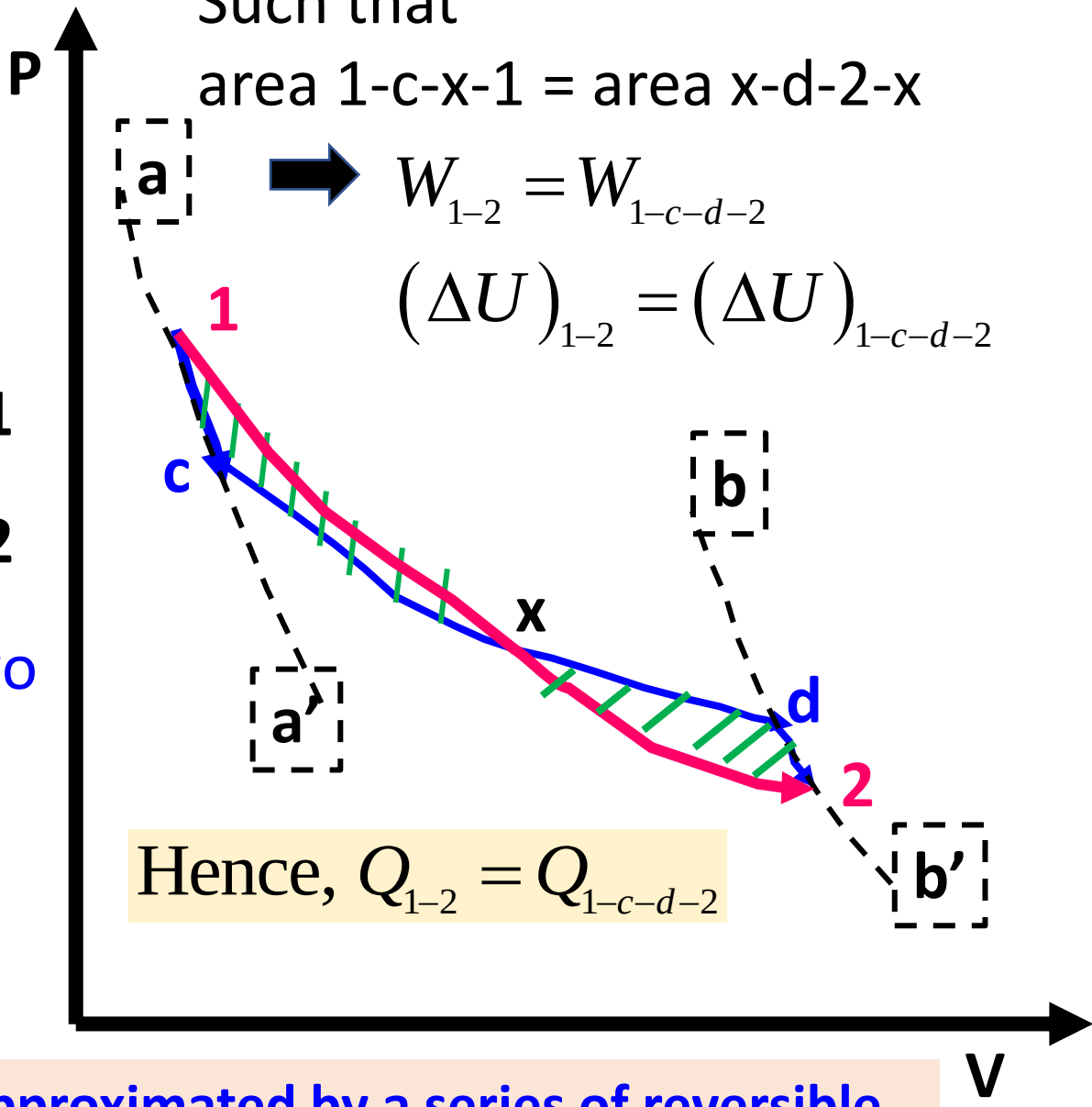
Let us draw two reversible isotherms and two reversible adiabatic processes connecting 1 and 2

1-c & d-2: Reversible adiabatic lines

c-d: Reversible isotherm connecting a-a' and b-b'

Such that

area 1-c-x-1 = area x-d-2-x



$$W_{1-2} = W_{1-c-d-2}$$

$$(\Delta U)_{1-2} = (\Delta U)_{1-c-d-2}$$

Hence, $Q_{1-2} = Q_{1-c-d-2}$

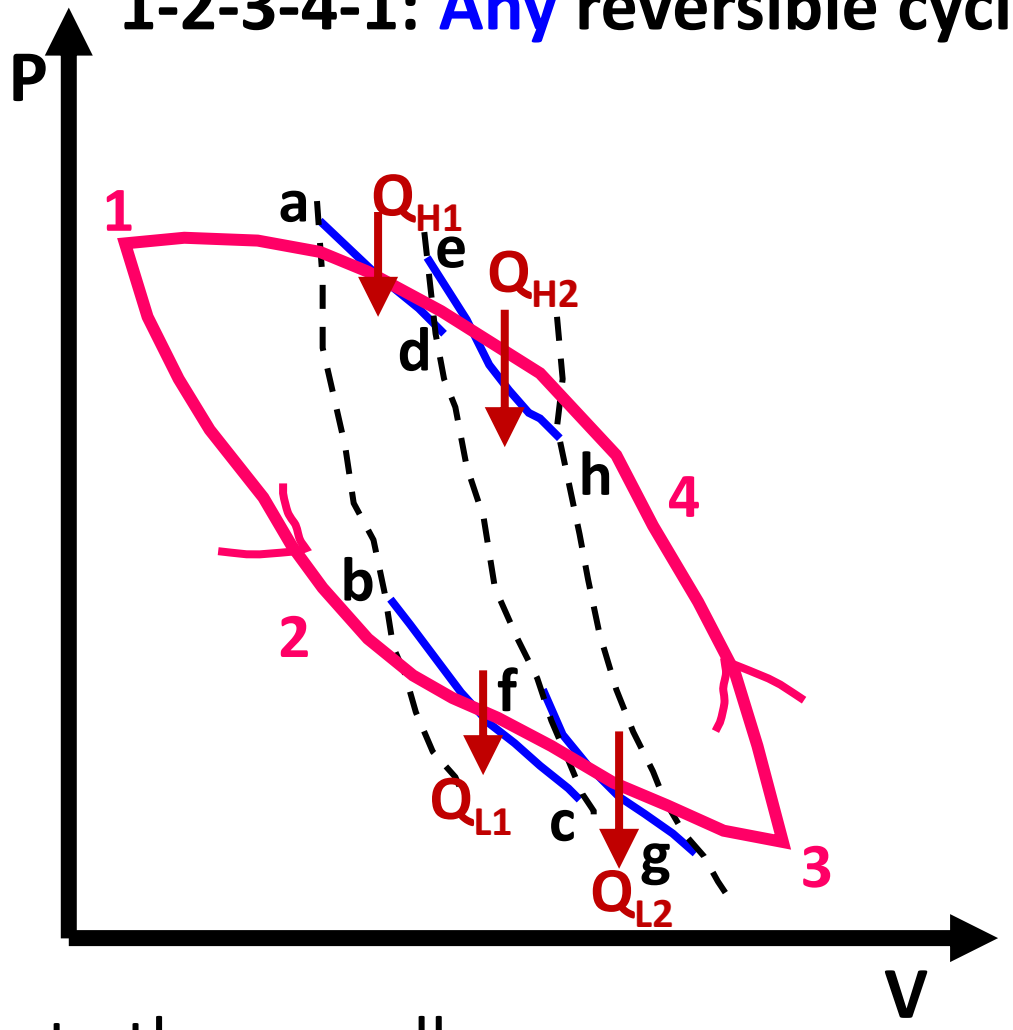
Thus, any reversible process can be approximated by a series of reversible isothermal and reversible adiabatic processes

Any reversible cycle

Let us consider any reversible cycle, i.e.

a-b, c-f-d-e & h-g: Reversible adiabatic lines

a-d, b-c, e-h & f-g: Reversible isotherms



- Consider cycles **a-b-c-d-a** and **e-f-g-h-e**

$$\frac{Q_{H1}}{T_{H1}} = \frac{Q_{L1}}{T_{L1}}, \qquad \frac{Q_{H2}}{T_{H2}} = \frac{Q_{L2}}{T_{L2}}$$

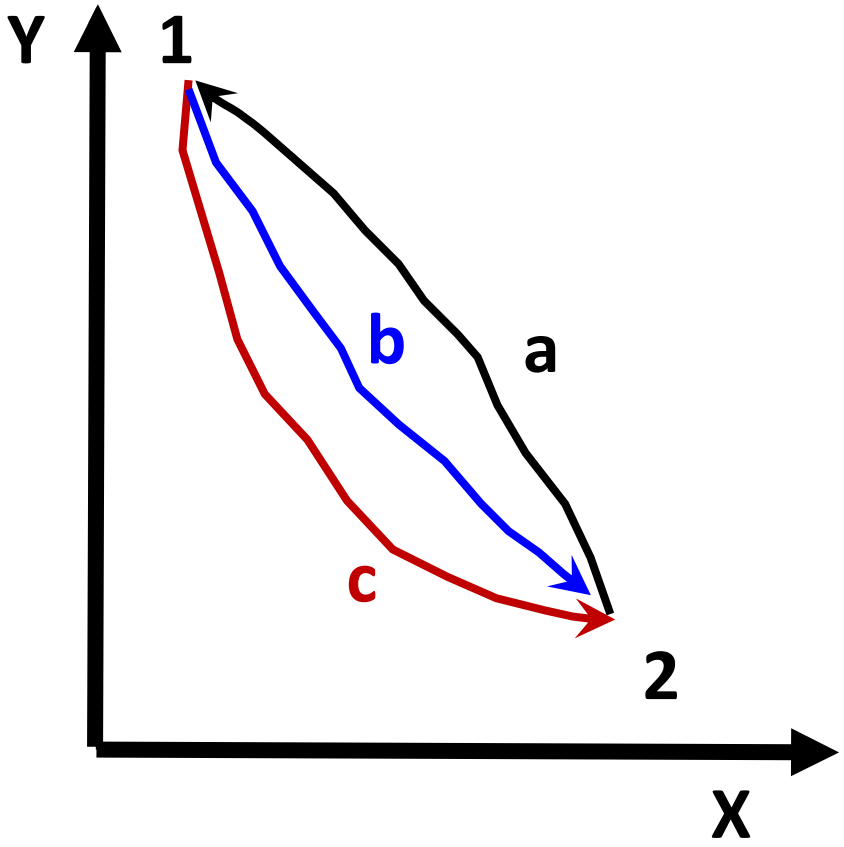
$$\frac{Q_{H1}}{T_{H1}} - \frac{Q_{L1}}{T_{L1}} = \frac{Q_{H2}}{T_{H2}} - \frac{Q_{L2}}{T_{L2}} = 0$$

- Extending this to all the small cycles that constitute the overall cycle 1-2-3-4

$$\sum \frac{Q}{T} = 0 \qquad \oint_{\text{rev cycle}} \frac{\delta Q}{T} = 0$$

Entropy: Path Independence

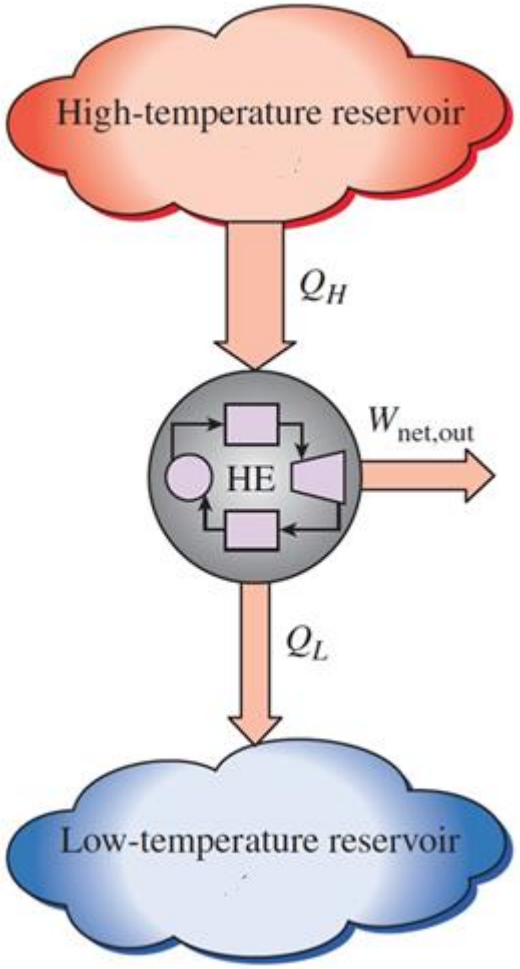
- a, b, c are reversible processes
- Consider the cycles 1-c-2-a-1 and 1-b-2-a-1
- $\oint \frac{\delta Q}{T} = \int_{1c}^{2c} \frac{\delta Q}{T} + \int_{2a}^{1a} \frac{\delta Q}{T} = \int_{1b}^{2b} \frac{\delta Q}{T} + \int_{2a}^{1a} \frac{\delta Q}{T} = 0$
- $\int_{1c}^{2c} \frac{\delta Q}{T} = \int_{1b}^{2b} \frac{\delta Q}{T}$
- Hence it is concluded that
- $\int_{rev} \frac{\delta Q}{T}$ = is the same for all reversible pathsand is the change in the property entropy
- $dS = \left(\frac{\delta Q}{T}\right)_{rev}$ and $S_2\text{-}S_1 = \int_1^2 \left(\frac{\delta Q}{T}\right)_{rev}$



Irreversible Heat Engine

Irreversible heat engines

The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two thermal reservoirs



Both these engines take the **same quantity of heat from high-temperature** thermal reservoir but have different net work

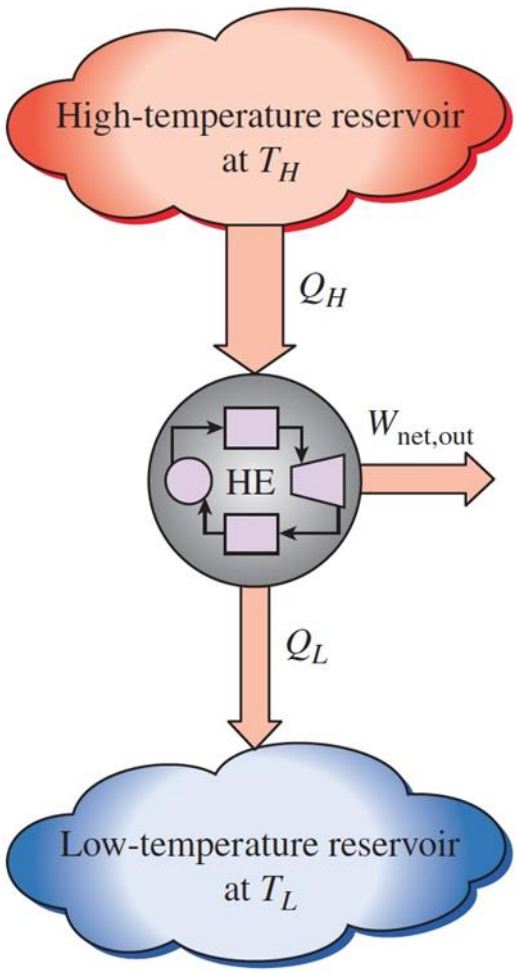
Let $\boldsymbol{W_{rev}}$ and $\boldsymbol{W_{irrev}}$ be the respective net work output from reversible and irreversible heat engines.

➔ Heat rejected $\boldsymbol{Q_{L,rev}}$ and $\boldsymbol{Q_{L,irrev}}$ are different.

$$\eta_{\text{irr HE}} < \eta_{\text{rev HE}} \Rightarrow \frac{Q_H - Q_{L,irrev}}{Q_H} < \frac{Q_H - Q_{L,rev}}{Q_H} \Rightarrow Q_{L,rev} < Q_{L,irrev}$$

Entropy change in irreversible process

Reversible heat engines



$$\frac{Q_{\text{rev}\,H}}{Q_{\text{rev}\,L}} = \frac{T_H}{T_L}$$

$$\frac{Q_{\text{rev}\,H}}{T_H} - \frac{Q_{\text{rev}\,L}}{T_L} = 0$$

$$\frac{Q_{\text{rev}\,H}}{T_H} + \left(\frac{-Q_{\text{rev}\,L}}{T_L}\right) = 0$$

$$\oint \left(\frac{\delta Q}{T}\right)_{\text{rev}} = 0$$

$$dS = \frac{\delta Q_{\text{rev}}}{T}$$

Irreversible heat engines

$$\oint \left(\frac{\delta Q}{T}\right)_{\text{irrev}} = \frac{Q_H}{T_H} - \frac{Q_{L,irrev}}{T_L}$$

$$= \left(\frac{Q_H}{T_H} - \frac{Q_{L,rev}}{T_L}\right) - \left(\frac{Q_{L,irrev}}{T_L} - \frac{Q_{L,rev}}{T_L}\right)$$

$$= -\left(\frac{Q_{L,irrev}}{T_L} - \frac{Q_{L,rev}}{T_L}\right)$$

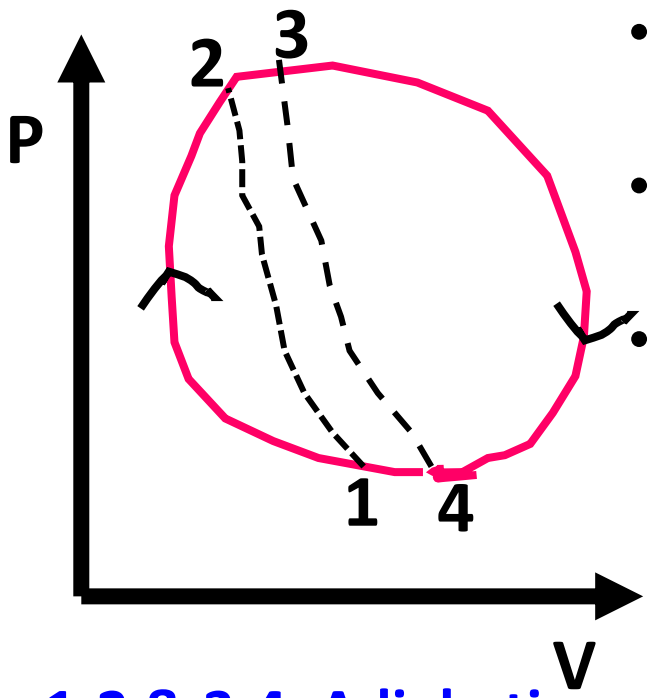
$$\oint \left(\frac{\delta Q}{T}\right)_{\text{irrev}} < 0 \qquad \because \quad Q_{L,rev} < Q_{L,irrev}$$

In general, $\oint \left(\frac{\delta Q}{T}\right) \le 0$ **Clausius Inequality**

Entropy change in any cycle

$$\oint \left(\frac{\delta Q}{T}\right) \leq 0$$

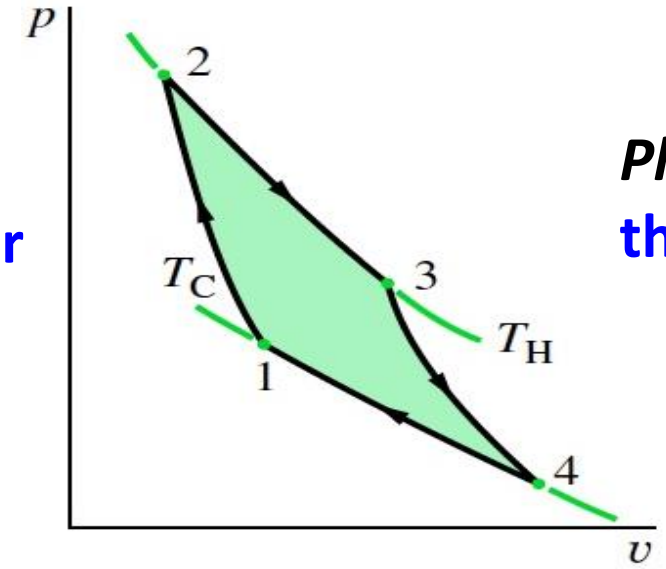
Clausius Inequality



- While deriving in the last slide, we have not considered difference in temperature between thermal reservoirs and the heat engine.
- Actually, there will be a finite temperature difference for heat transfer.
- Hence, in this cyclic integral for an irreversible heat engine, strictly speaking, **we should consider the actual temperature of the heat engine** rather than the temperature of the thermal reservoir.

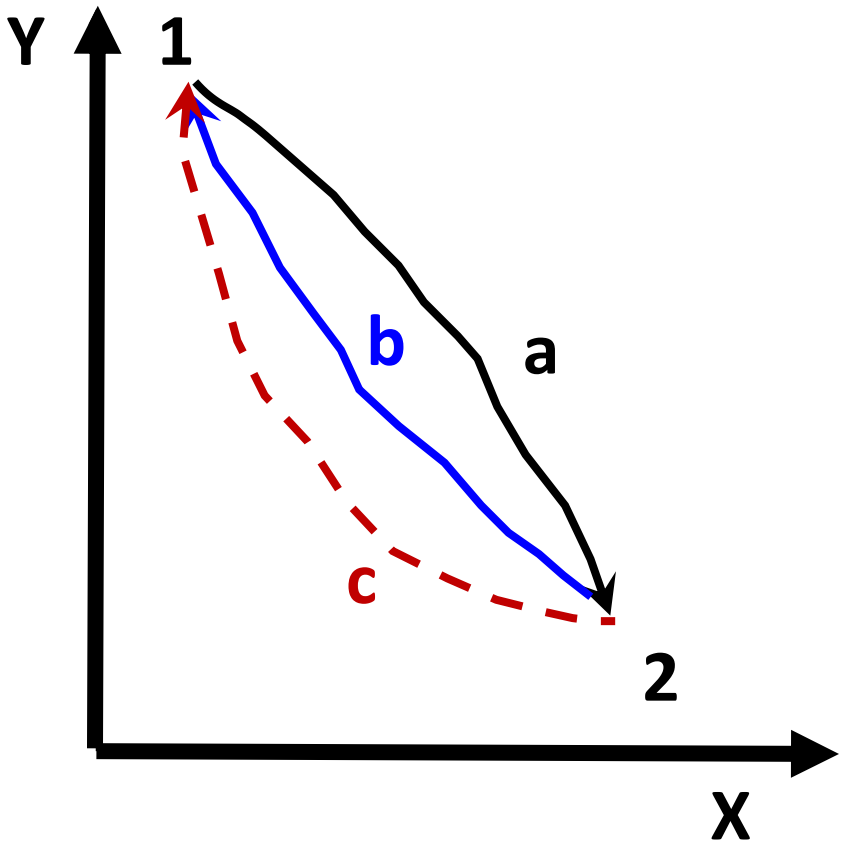
1-2 & 3-4: Adiabatic

2-3 & 4-1: Heat Transfer



Please note: **T_C and T_H are temperatures of the working fluid**

Entropy change in any process



1-a-2: Reversible process

2-b-1: Reversible process

2-c-1: Irreversible process

For the reversible cycle 1-a-2-b-1,

$$\oint \left(\frac{\delta Q}{T}\right)_{\text{rev}} = 0 \quad \Rightarrow \quad \int_1^2 \left(\frac{\delta Q}{T}\right)_a + \int_2^1 \left(\frac{\delta Q}{T}\right)_b = 0$$

For the irreversible cycle 1-a-2-c-1,

$$\oint \left(\frac{\delta Q}{T}\right)_{\text{rev}} < 0 \quad \Rightarrow \quad \int_1^2 \left(\frac{\delta Q}{T}\right)_a + \int_2^1 \left(\frac{\delta Q}{T}\right)_c < 0$$

$$\int_1^2 \left(\frac{\delta Q}{T}\right)_a + \int_2^1 \left(\frac{\delta Q}{T}\right)_c < \int_1^2 \left(\frac{\delta Q}{T}\right)_a + \int_2^1 \left(\frac{\delta Q}{T}\right)_b$$

$$\int_2^1 \left(\frac{\delta Q}{T}\right)_c < \int_2^1 \left(\frac{\delta Q}{T}\right)_b < \Delta S \quad \Rightarrow \quad dS > \left(\frac{\delta Q}{T}\right)_{\text{irrev}} \quad \Rightarrow \quad dS \geq \left(\frac{\delta Q}{T}\right)$$

Entropy changes during a process

- In general, $dS \geq \frac{\delta Q}{T}$
- Heat transfer to a system ➔ increase in the entropy.
- Heat transfer from a system ➔ decrease in entropy.
- Work on/by the system: **as long as it is reversible** (does not involve friction, flow of current through a resistor etc.) **will not change the entropy**.
- Work on/by the system: and **involves dissipative effects (**like friction, flow of current through a resistor etc.) will **increase entropy**.
- There could be combined effects of heat and work on changes in entropy
- Any irreversibility like heat transfer through a finite temperature difference, friction and other dissipative effects will only increase in entropy
- **So in any process $\Delta S_{system} = \Delta S_{due\ to\ heat\ transfer} + \delta S_{production}$**
- **$\delta S_{production}$ = Entropy generated due to dissipative effect (irreversibility) ≥ 0**

Entropy Generation

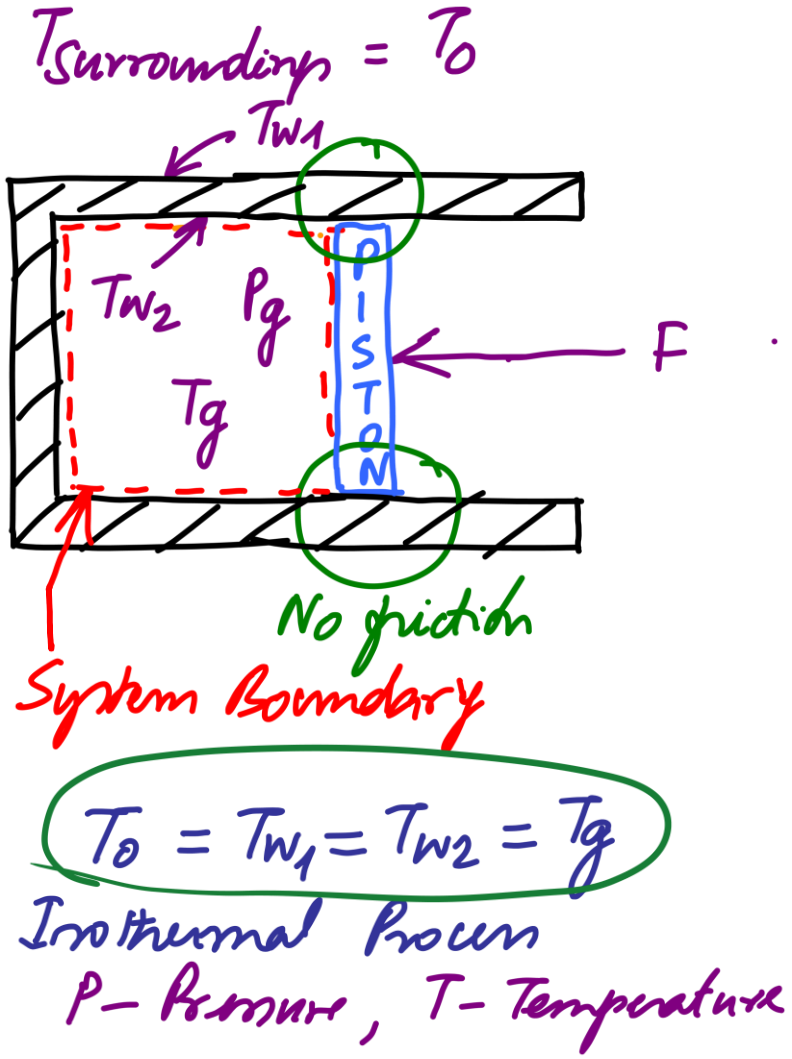
$$dS = \frac{\delta Q}{T} + \delta S_{gen} \qquad \delta S_{gen} \geq 0$$

$\delta Sgen$ is the **entropy generated due to irreversibility** occurring during the process **within the system** like friction, internal heat transfer across finite temperature difference, electrical resistance, unresisted expansion etc.

Additionally, there can be irreversibility occurring external to the system, For example in transferring heat to the system through a finite temperature difference or work transfer with friction etc.

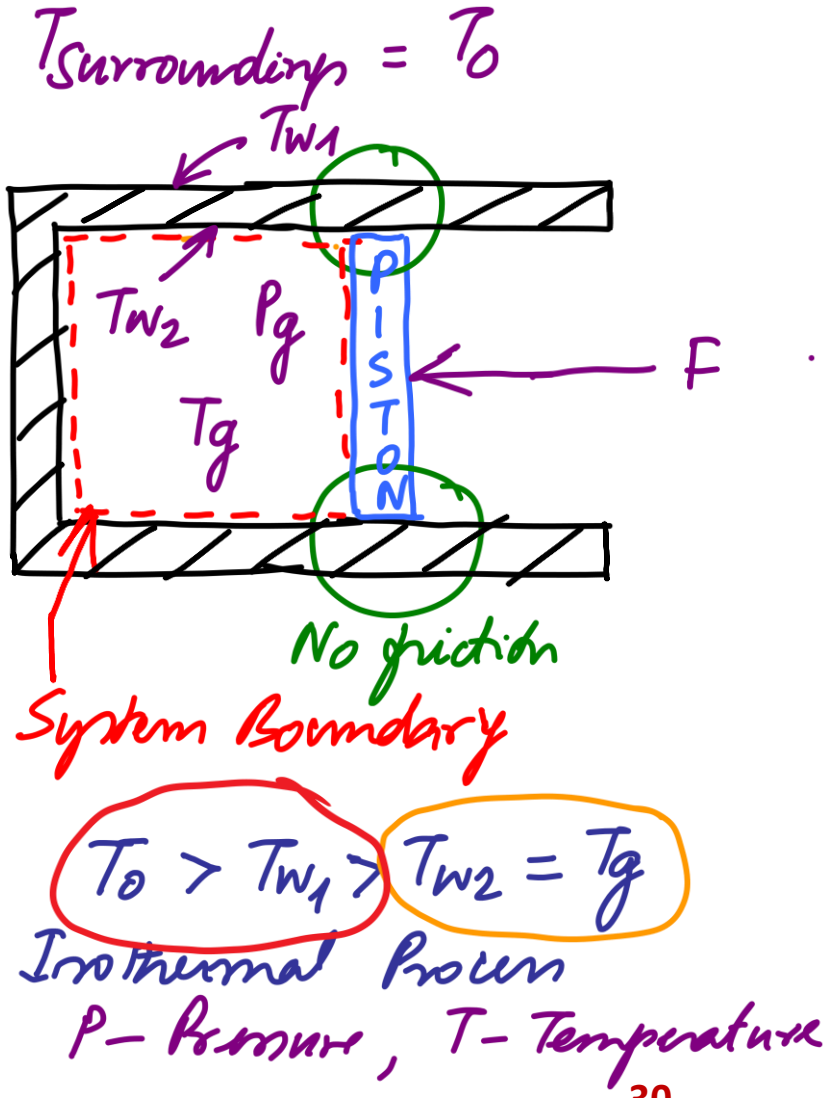
Internal and External irreversibility

- Consider the following system (piston and the gas inside a cylinder).
- There is no friction and the process in slow and there is no temperature difference between it and the surroundings
- This process is **completely reversible** as there is no heat transfer across a finite temperature difference, no dissipative effects like friction etc.



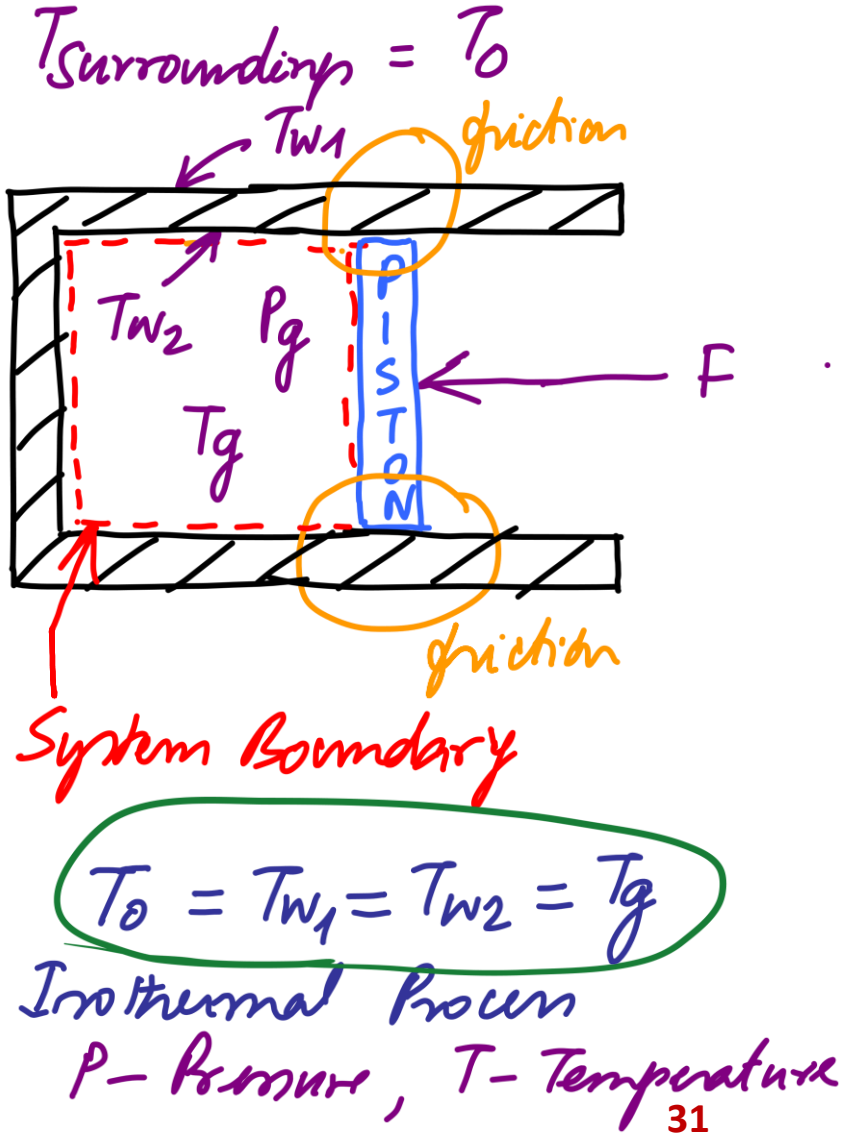
Internal and External irreversibility

- In the same system if there is a temperature gradient in the cylinder wall
  - But the inner wall temperature is always extremely close to the temperature of the gas in the cylinder
  - There are also no dissipative effects like friction.
- The process is still **internally reversible**
- But there are irreversibilities outside the system.
- The process **is externally irreversible**
- Generally, we have been dealing with internally reversible processes that may or may not be externally reversible
- We have been focussing on the system and not what was outside it



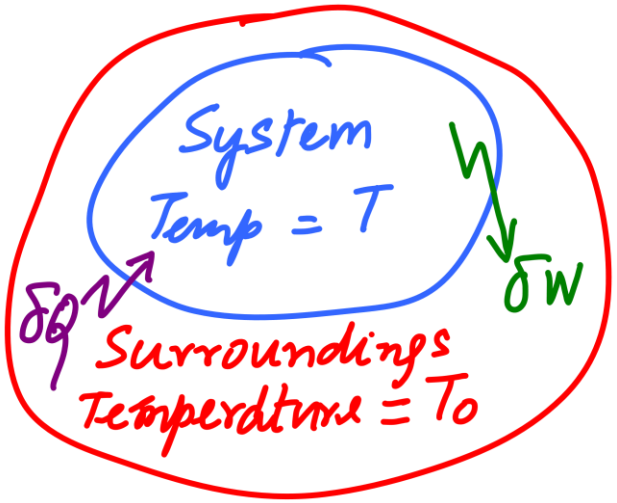
Internal and External irreversibility

- A process like the one shown on the right that has friction between the piston and cylinder
  - But no heat interaction with the surroundings or heat interaction only across an infinitesimally small temperature difference
- This is **internally irreversible**
- **Externally reversible**

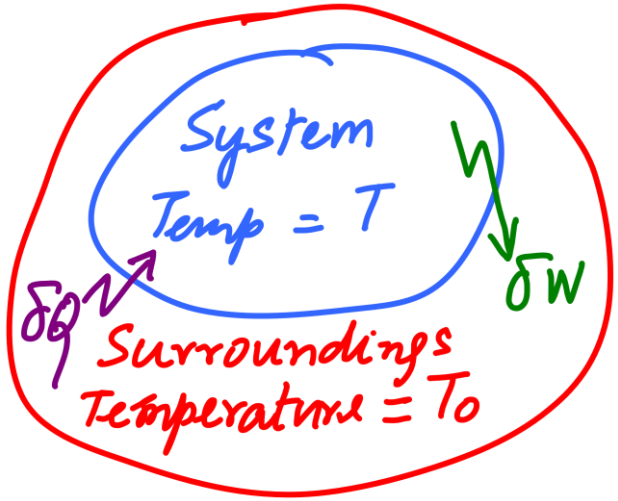


Principle of Increase of Entropy

- So far we have been dealing with systems and thus have been considering internally reversible and internally irreversible processes.
- Entropy of a system can be increased by entropy generation or by heat transfer to the system but can only be decreased by heat transfer from the system
- Let us now consider a system and its surroundings
- $dS_{sys} \geq \frac{\delta Q}{T}$,
- for the surroundings, heat transfer is negative
- $dS_{surr} = \frac{-\delta Q}{T_0}$ $hence\ dSnet = dSs_{ys} + dSs_{urr}$



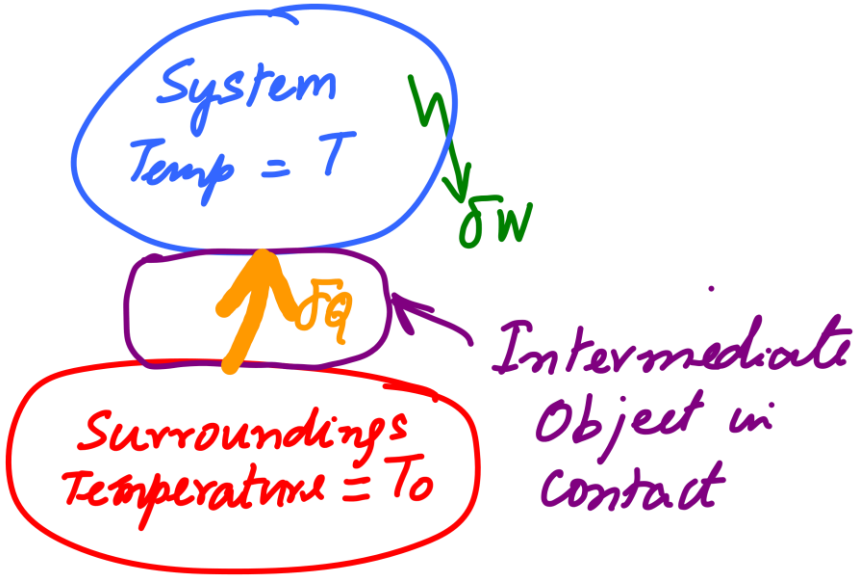
Principle of Increase of Entropy



- $dS_{net} \geq \left(\frac{\delta Q}{T} - \frac{\delta Q}{T_0}\right)$
- $dS_{net} \geq \delta Q \left(\frac{1}{T} - \frac{1}{T_0}\right)$,
- $\delta Q \left(\frac{1}{T} - \frac{1}{T_0}\right) > 0$
- $Since\ T_0 > T,\ \ dSnet > 0\ \ i.e\ dS_{sys} + dS_{surr} > 0$
- If the $T_0 < T,\ \ \ Q\ will\ be\ negative\ and\ we\ will\ get\ the\ same\ result$
- $The\ term\ dS_{net}\ is\ the\ total\ entropy\ generation\ in\ this\ process$
- The term on the right hand side is the entropy generation external to the system due to the heat transfer across (T_0-T) finite temperature difference
- It is the external entropy generation in addition to the internal entropy generation with in the system

Principle of Increase of Entropy

- To understand this external entropy generation let us consider the figure on the right which is a better representation of the previous figure
- Here the intermediate object that could be anything is transferring the heat, and it is at the contact temperatures on each side
- $dS_{object} = \left(\frac{\delta Q}{T_0} - \frac{\delta Q}{T}\right) + dS_{gen,\ object} = 0$
  - Because the intermediate object does not undergo any change in state
- $dS_{gen,\ object} = \left(\frac{\delta Q}{T} - \frac{\delta Q}{T_0}\right) > 0 \ as \ T < T_0$



Principle of Increase of Entropy

- In the previous case there was entropy generation in the system and also in the intermediate member which is a part of the surroundings.

- The entropy generation in the system is because of what takes place inside the system like internal heat transfers across finite temperature differences and other dissipative effects

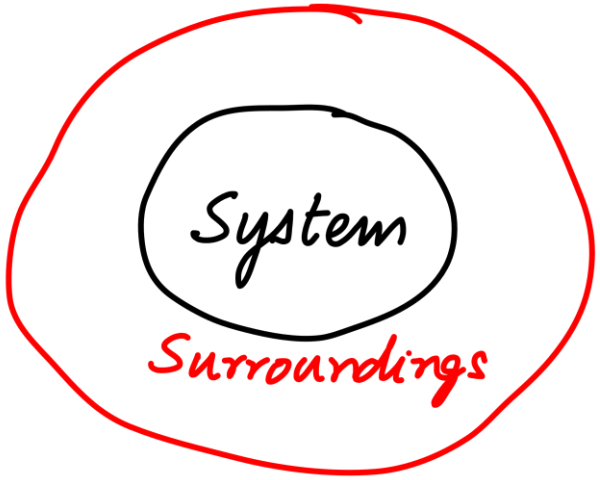
- Similarly, there can be such effects in the surroundings

- Thus $\boldsymbol{dS_{sys} + dS_{surr} > 0}$

- The above equation is called the **principle of increase in entropy**

Principle of Increase of Entropy

- If we take an isolated system
  - One that can have no heat or work interaction (no interaction) with the surroundings
  - $dS \geq \frac{\delta Q}{T}$
  - $Q = 0 \; for \; an \; isolated \; system$
  - $dS_{isolated\ system} \geq 0$
- Since an isolated system is also an adiabatic system
  - $dS_{adiabatic\ system} \geq 0$
- The entropy of an isolated system always increases and in the limiting case of reversible processes becomes constant with respect to time.
  - If the entropy of an isolated system is constant with time, then only reversible processes are occurring in the system
  - If the entropy of an isolated system increases with time, then irreversible processes are occurring in the system
  - The entropy of an isolated system can never decrease.

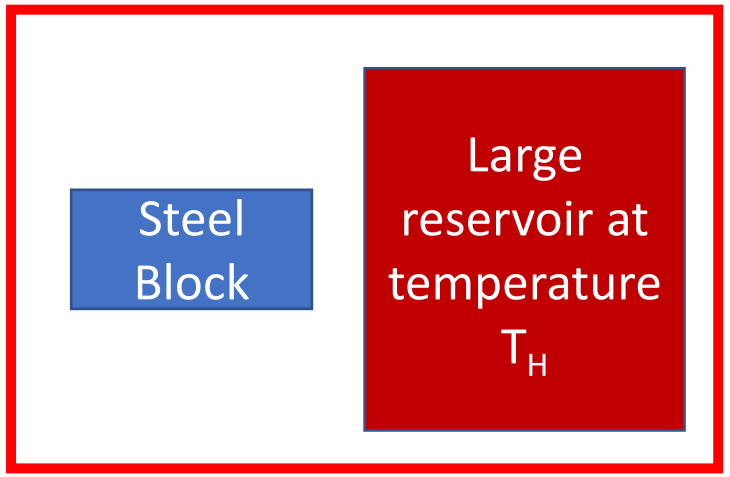


System + Surroundings = Universe

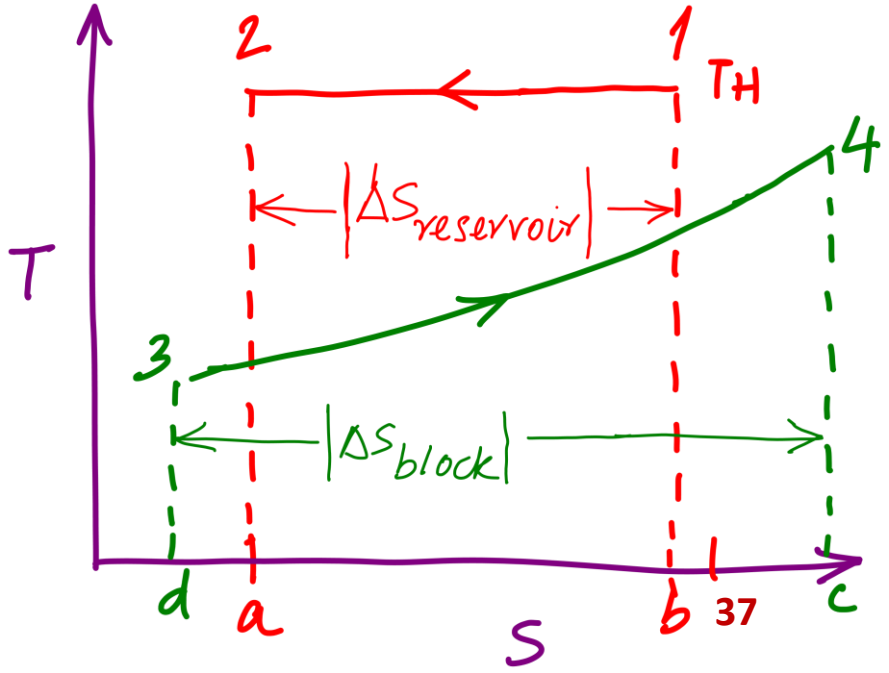
Universe is an isolated System
No heat or work interactions

Principle of Increase of Entropy

- Consider the isolated system shown
- The block increases its temperature from T_3 to T_4 by heat transfer from the reservoir at a higher temperature T_H
- The heat transfer processes in each of these subsystems is reversible internally
- The processes are shown in the graph
- The heat transfers and the area under the curves is the same
  - Area (1-2-a-b-1) = Area (3-4-c-d-3)
- The entropy increase of the steel block is higher that the entropy decrease of the reservoir.
- $dS_{isolated\ System}$ >0



Isolated System

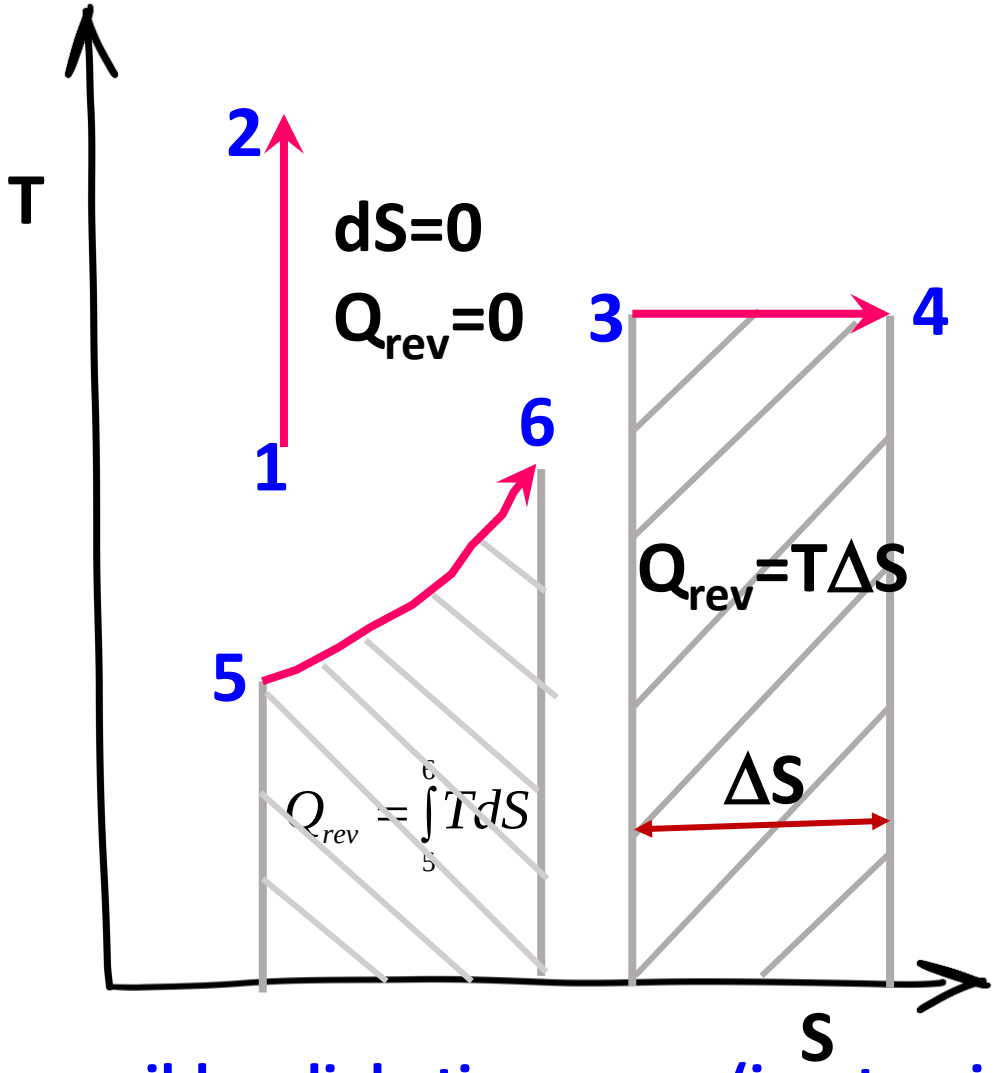


Comments on Principle of Increase of Entropy

- The principle of increase in entropy indicates that
  - All the processes that can take place will only increase the entropy of the system plus surroundings put together or in the limit keep the sum of their entropies constant.
  - In other words the system and the surroundings cannot be returned to their original states --- this can never occur as it will involve a decrease in their total entropy.
  - When there is no process occurring i.e. stability is reached, entropy of the system plus surroundings will become constant
- Thus this principle indicates the direction in which processes like temperature change of coffee left on the table, combustion in a engine, flow of water etc. can occur.

Use of entropy as a coordinate to represent processes

- We have seen that $dS = \frac{\delta Q_{rev}}{T}$
- Hence the heat transfer in a reversible process, $Q_{rev} = \int_1^2 T dS$
- Similar to work in a reversible process, $W_{rev} = \int_1^2 p dV$
- Hence, in the case of **reversible processes** the **area under the T-S diagram** represents the **heat transfer**
- The area enclosed by a **reversible cycle** on the T-S coordinates indicates the **net heat transfer for the cycle which is also the net work done**
- Hence T-S diagrams are widely used to analyse thermodynamic processes in engineering applications



- **1-2: Reversible adiabatic process (isentropic)**
- **3-4: Reversible isothermal process**
- **5-6: General reversible process**

Use of entropy as a coordinate to represent processes…

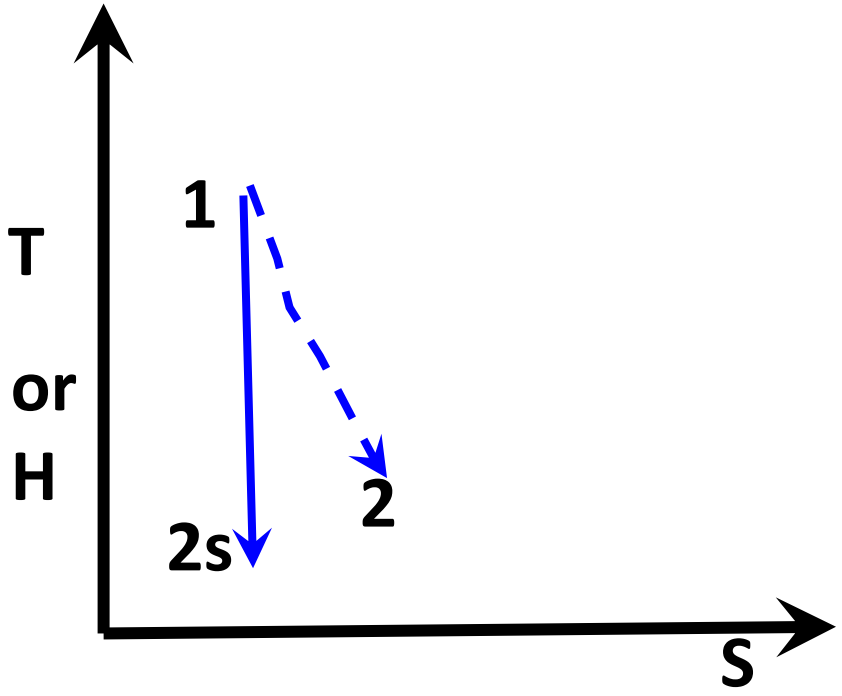
- **Reversible adiabatic process** ➔ **$Q = 0 = \delta S_{gen}$**

$$dS = \frac{\delta Q}{T} + \delta S_{gen} \quad \Rightarrow \quad dS = 0$$

Isentropic process: 1-2s

While handling flows through nozzles, compressors and turbines the flow is often taken to be reversible and adiabatic

Here it is useful to represent such processes on the T-S or H-S coordinates



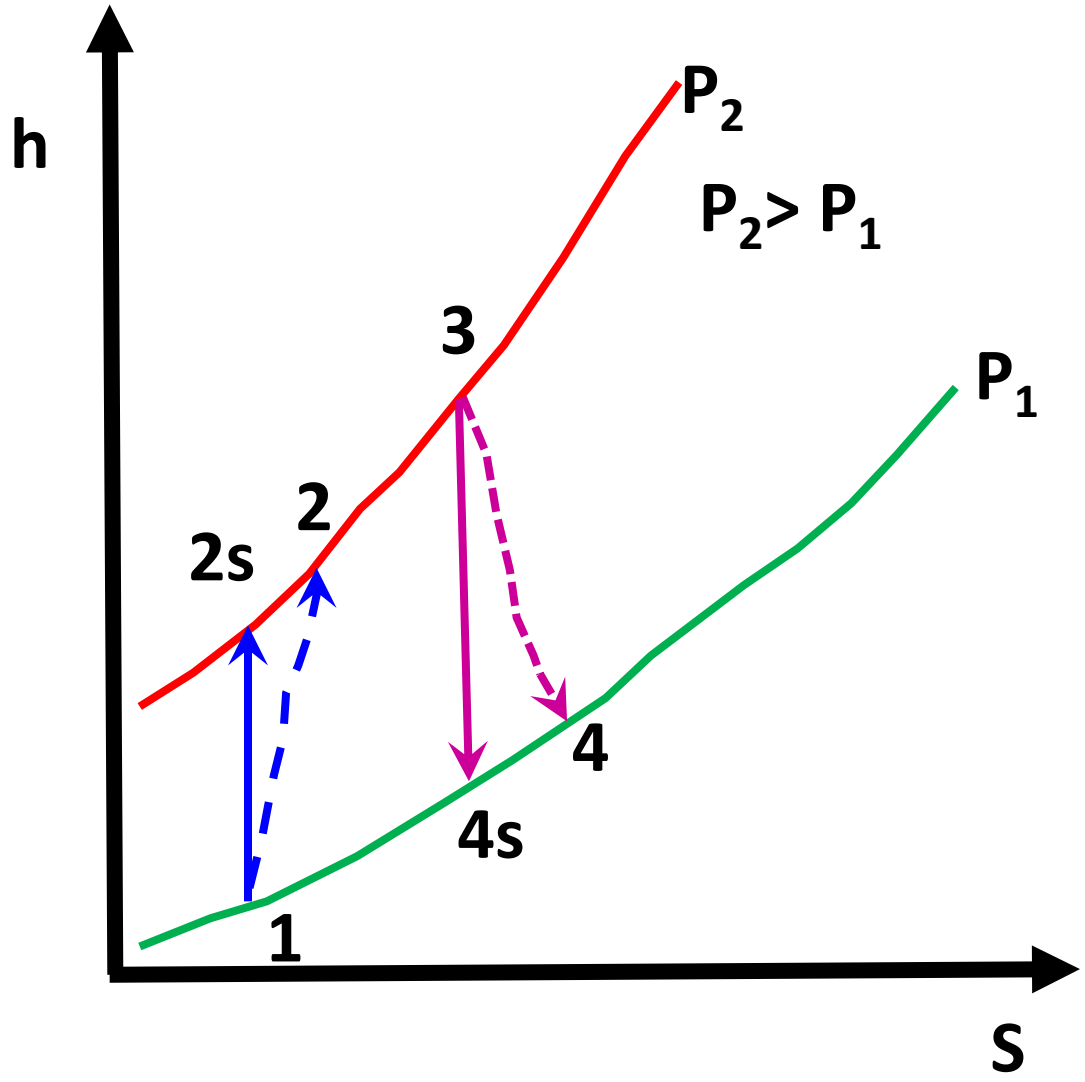
- **Irreversible adiabatic process** ➔ **$Q = 0$; $\delta S_{gen} > 0$**

Process: 1-2

Reversible and irreversible adiabatic processes shown on h-s coordinates

- 1-2s reversible adiabatic compression ΔS=0
- 1-2 irreversible adiabatic compression ΔS>0

- 3-4s reversible adiabatic expansion ΔS=0
- 3-4 irreversible adiabatic expansion ΔS>0

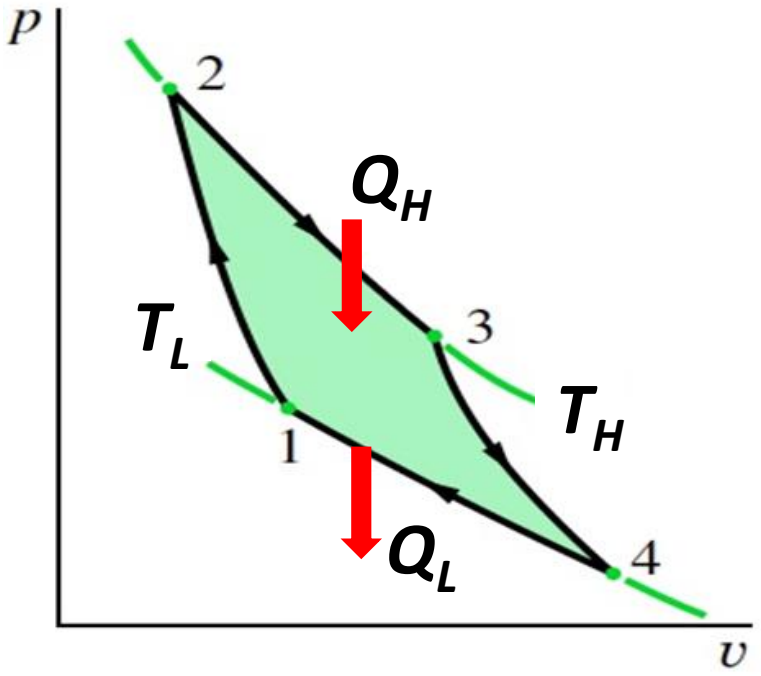


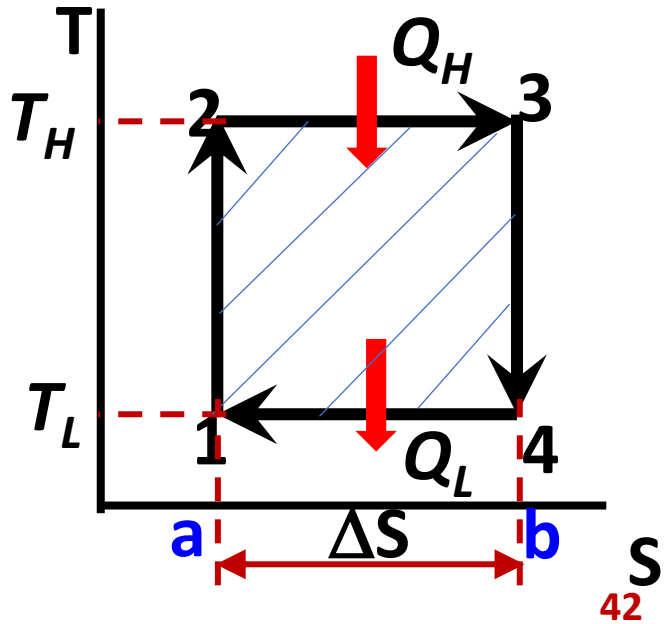
Carnot cycle on the T-S coordinates

- ***All are reversible processes*** → $Q \equiv Q_{rev}, \delta S_{gen}=0$
- $dS = \frac{\delta Q}{T}$
- $Process\ 1-2, \quad S_1 - S_2 = 0$
- $Process\ 2-3, S_3 - S_2 = \int_2^3 \frac{\delta Q}{T} = \frac{Q_H}{T_H} = \Delta S$
  - $Q_H = T_H * \Delta S$ = Area 2-3-b-a-2
- $Process\ 3-4, S_4 - S_3 = 0$
- $Process\ 4-1, S_1 - S_4 = \int_4^1 \frac{\delta Q}{T} = \frac{Q_L}{T_L} = -\Delta S$
  - $Q_L = -T_L * \Delta S$ = Area 1-4-b-a-1
- $\oint \Delta Q = QH + QL = (T_H - T_L)\ \Delta S = W_{cycle}$

$W_{cycle} = Q_{cycle}$ = (Area 2-3-b-a-2) – (Area 1-4-b-a-1)

i.e. Area 1-2-3-4-1

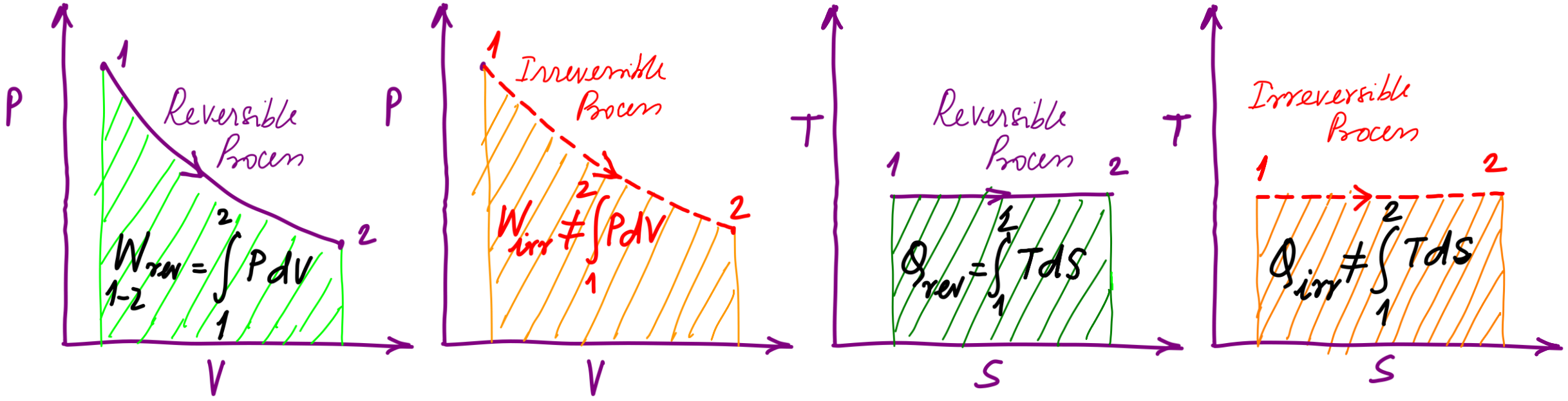




Remarks

- The area under the P-V curve will represent work **only** in a reversible process
- The area under the T-S curve will represent heat transfer **only** in a reversible process
- $S_2 - S_1 = \int_1^2 dS + S_{gen}$
- The entropy of a system can be increased by one of the two ways
  - Heat Transfer
  - Irreversible process (entropy generation is positive)
- Entropy can be decreased only by heat transfer (entropy generation is positive)
- Entropy is generated only in an irreversible process but a reversible process can have entropy increase due to heat transfer even though there is no entropy generation

To summarize



Reversible processes are represented by solid lines

Irreversible processes are represented by dashed lines

Entropy calculations for various processes and substances

- The entropy change between any two states 1,2 can be obtained by using **any reversible process** that connects these two states.
  - $dS = \frac{\delta Q_{rev}}{T}, \Rightarrow TdS = \delta Q_{rev}$

The first law of thermodynamics for a **simple compressible system** (involving **displacement, p-dV, work** as the only work) and undergoing **reversible process** and can be written as

$$\delta Q_{rev} = dU + pdV \quad \Rightarrow \quad \boldsymbol{TdS = dU + PdV}$$

We know that H = U + PV ➔ dH = dU + pdV + VdP ➡ $\boldsymbol{TdS = dH - VdP}$

The above relations connect **different properties.**
So, these can be used for any process between the two states, **both reversible** and **irreversible**.

Since entropy is a property, even for an **irreversible process** between two states, **entropy change** can be computed by assuming **any reversible process between the same two states**.

Entropy change for an ideal gas undergoing a process

Using the First TdS relationship

- $TdS = dU + PdV = mC_v dT + \frac{mRT}{V} dV$
- $dS = m\left[C_v . \frac{dT}{T} + R . \frac{dV}{V}\right]$
- $\Delta S = m\left[C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right)\right]$

Using the Second TdS relationship

- $TdS = dH - VdP = mC_p dT - \frac{mRT}{P} d\mathrm{P}$
- $dS = m\left[C_p . \frac{dT}{T} - R . \frac{dP}{P}\right]$
- $\Delta S = m\left[C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)\right]$

Entropy change for an ideal gas undergoing a process

From the First TdS relationship

$$\Delta S = m\left[C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right)\right]$$

- For an ideal gas undergoing a reversible adiabatic process
- $\Delta S = m\left[C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right)\right] = 0$
- This leads to $\left(\frac{T_2}{T_1}\right)^{C_v} = \left(\frac{V_1}{V_2}\right)^{R}$
- $\left(\frac{T_2}{T_1}\right) = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$
- It follows that for a reversible adiabatic process

$\left(\frac{P_2}{P_1}\right) = \left(\frac{V_1}{V_2}\right)^{\gamma}$ or PV^{γ} = constant, for an isentropic process, S=Constant

Entropy change for an ideal gas undergoing a process

From the First TdS relationship

$$\Delta S = m\left[C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)\right]$$

- For an ideal gas undergoing a reversible adiabatic process this can be written as
- $\Delta S = 0 = m\left[C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)\right]$

- This leads to $\left(\frac{T_2}{T_1}\right)^{C_p} = \left(\frac{P_2}{P_1}\right)^{R}$

- $\left(\frac{T_2}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$ Where $\gamma = \frac{C_p}{C_v}$ and $C_p = \frac{R\gamma}{\gamma-1}$ $and C_v = \frac{R}{\gamma-1}$

Entropy changes in solids and liquids

- In solids and liquids when the temperature changes, the accompanying volume changes are negligible.
  - This is not true for gases wherein volume changes are not negligible and we took this into account in our earlier equations
- Thus in the case of solids and liquids C_p and C_v values are almost the same and can be considered to be equal
- We take $C_p = C_v = C$
- The first TdS relationship gets simplified to
- $TdS = dU + pdV$
- TdS $= dU = mCdT$

$$dS = mC\left(\frac{dT}{T}\right) \; which \; leads \; to \; \Delta S = m.C.ln\left(\frac{T_2}{T_1}\right)$$

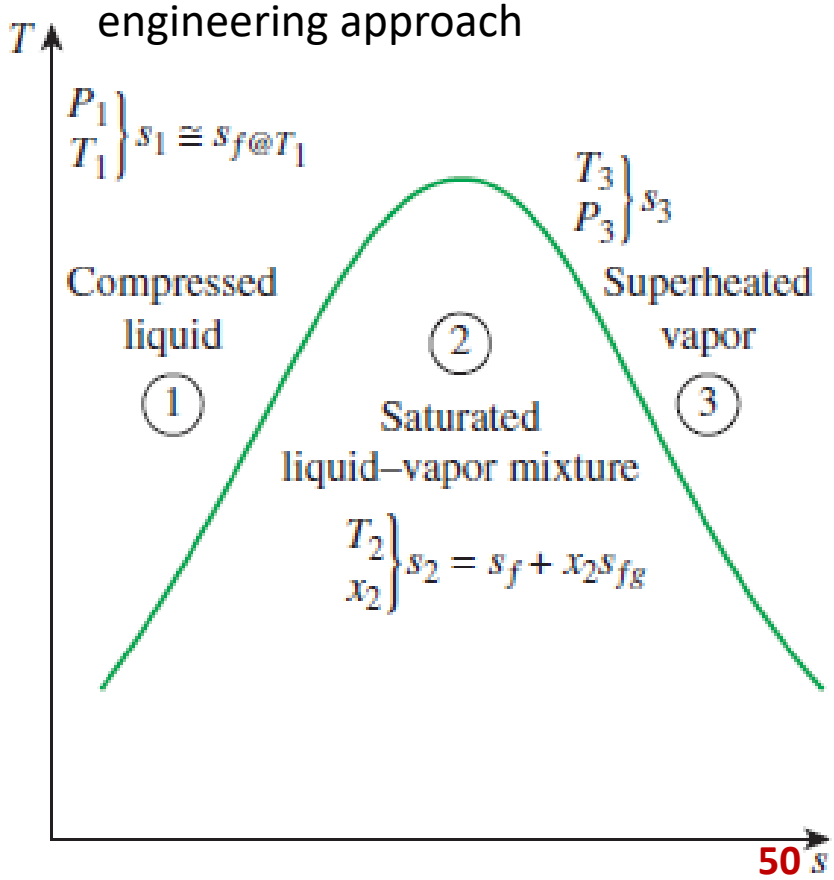
Entropy changes – Steam Table

T , °C	p, bar	v_f, m^3/kg	v_g, m^3/kg	u_f, kJ/kg	u_g, kJ/kg	h_f, kJ/kg	h_g, kJ/kg	s_f, kJ/kg.K	s_g, kJ/kg.K
0.01	0.0061	1.0002×10^{-3}	206.1	0.01	2376	0.01	2501	0.0	9.156
5.0	0.0087	1.0001×10^{-3}	147.1	21.00	2383	21.00	2511	0.0762	9.026
10.0	0.0123	1.0001×10^{-3}	106.4	42.01	2389	42.01	2520	0.151	8.901

Pressure p = 2.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg.K
150.0	0.960	2577	2769	7.280
200.0	1.080	2654	2871	7.507
250.0	1.199	2731	2971	7.709

- $s = s_f + x s_{fg}$
- $s_{fg} = s_g - s_f$
- $s = (1 - x)s_f + x s_g$
- $s_{comp.liq.}(T, P) = s_{sat.liq.}(T)$

Çengel and Boles: Thermodynamics – an engineering approach



Limits for heat transfer in irreversible processes

- Consider a simple compressible system (SCS) where the **only work done** is displacement (p-dV) work. For this system undergoing **reversible process**,

$$\delta W_{rev} = P\,dV \quad \text{and} \quad \delta Q_{rev} = T\,dS$$

- Now consider an irreversible process for this SCS ➔ $\delta W_{irr} \neq P\,dV$ & $\delta Q_{irr} \neq T\,dS$
- However, for any process $dS = \frac{\delta Q}{T} + \delta S_{gen}$

Therefore, for **an irreversible process** $\delta Q_{irr} = T\,dS - T\delta S_{gen}$

δS_{gen}

$= 0$ **for a reversible process and**

> 0 **for an irreversible process**

Limits for work in irreversible processes

We can also write: $\delta \mathbf{Q_{irr}} = \mathbf{dU} + \delta \boldsymbol{W_{irr}}$

T dS = dU + P dV is valid for all processes as it is a relationship between properties

Between any two states, *dU* is the same irrespective of the process

$$\delta \boldsymbol{Q_{irr}} = (\boldsymbol{T\ dS - P\ dV}) + \delta \boldsymbol{W_{irr}}$$

Also $\delta \mathbf{Q_{irr}} = \boldsymbol{T\ dS\ - T\delta S_{gen}}$

$$\cancel{\boldsymbol{T\ dS}}\ \boldsymbol{- T\ \delta S_{gen}} = \cancel{\boldsymbol{T\ dS}} \boldsymbol{- P\ dV} + \delta \boldsymbol{W_{irr}}$$

➡

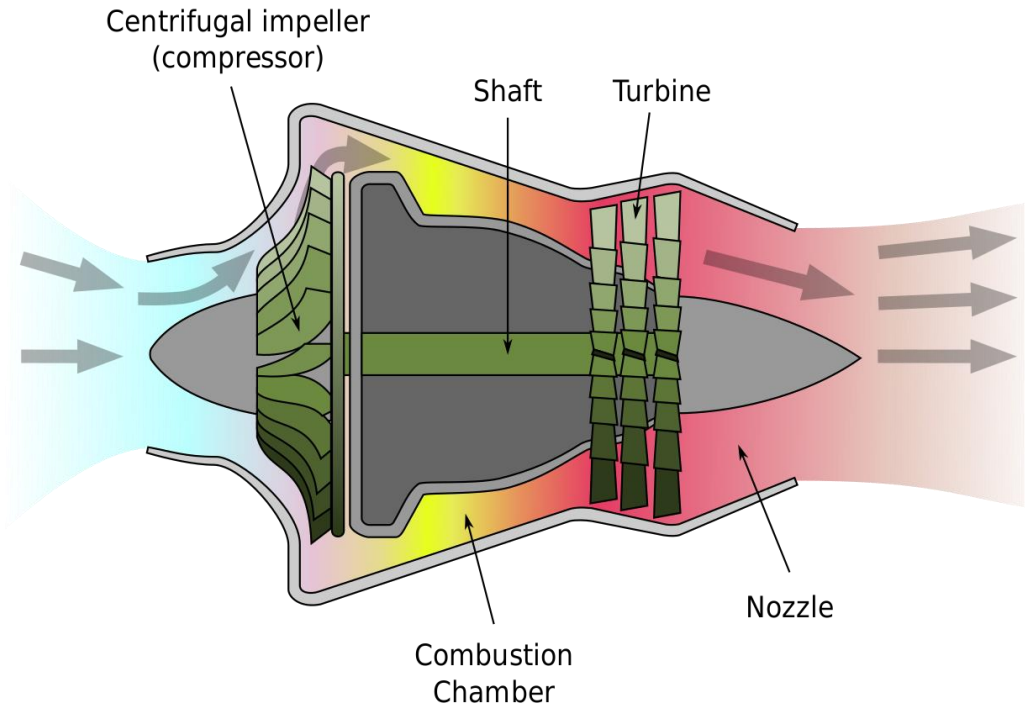
$$\delta \boldsymbol{W_{irr}} = \boldsymbol{P\ dV - T\delta S_{gen}}$$

- The quantity **T δS_{gen}** is often called **lost work**.
- It is not the energy lost or real work that is lost → it is the **lost opportunity to extract work** i.e. the **work** that could have been **extracted** for the **same P dV**.
- **Example:** From the same initial conditions (P, V) an irreversible adiabatic process will take to a different end state as compared to a reversible adiabatic process. If the end volumes are the same the end pressures will not be the same.

Isentropic efficiencies of some important steady flow devices

- Irreversibilities associated with any process cannot be eliminated but can be minimised
  - Irreversibilities downgrade the performance of devices
  - Result in more work input
  - Reduced work output
- We need a figure of merit to compare how good the performance of a device is with respect to the maximum that is achievable to that development becomes meaningful
- We can compare the actual performance of heat engines, refrigerators and heat pumps with a standard – Carnot Cycle
  - We can say that we have reached our goal of highest figure of merit if we reach Carnot Efficiency or Carnot COP

Jet Engine showing Compressor, Turbine and Nozzle

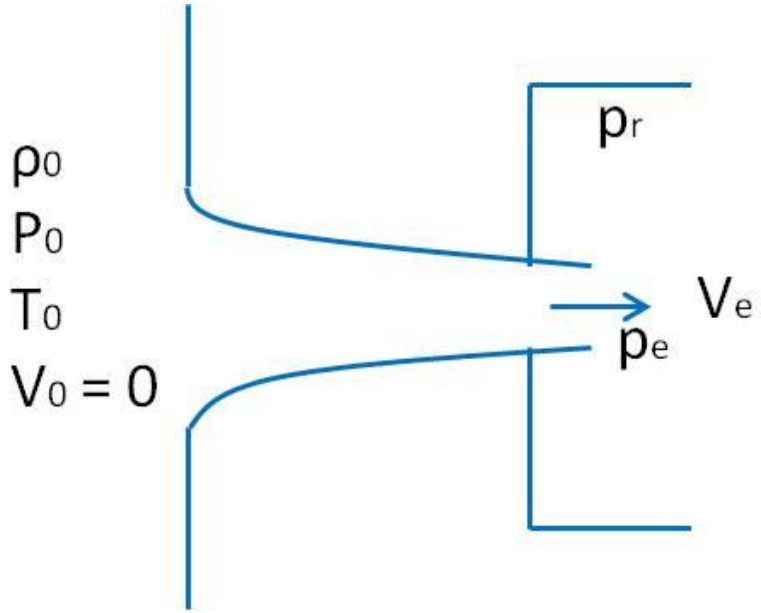


A Jet Engine showing Compressor, Turbine and Nozzle
https://commons.wikimedia.org/w/index.php?title=File:Turbojet_operation-centrifugal_flow-en.svg&oldid=464602741.



https://commons.wikimedia.org/w/index.php?title=File:Compressor_wheel_Napier_NA357.JPG&oldid=455070011.

Nozzles



Convergent Nozzle
https://commons.wikimedia.org/w/index.php?title=File:A_Converging_Nozzle.jpg&oldid=492524066.

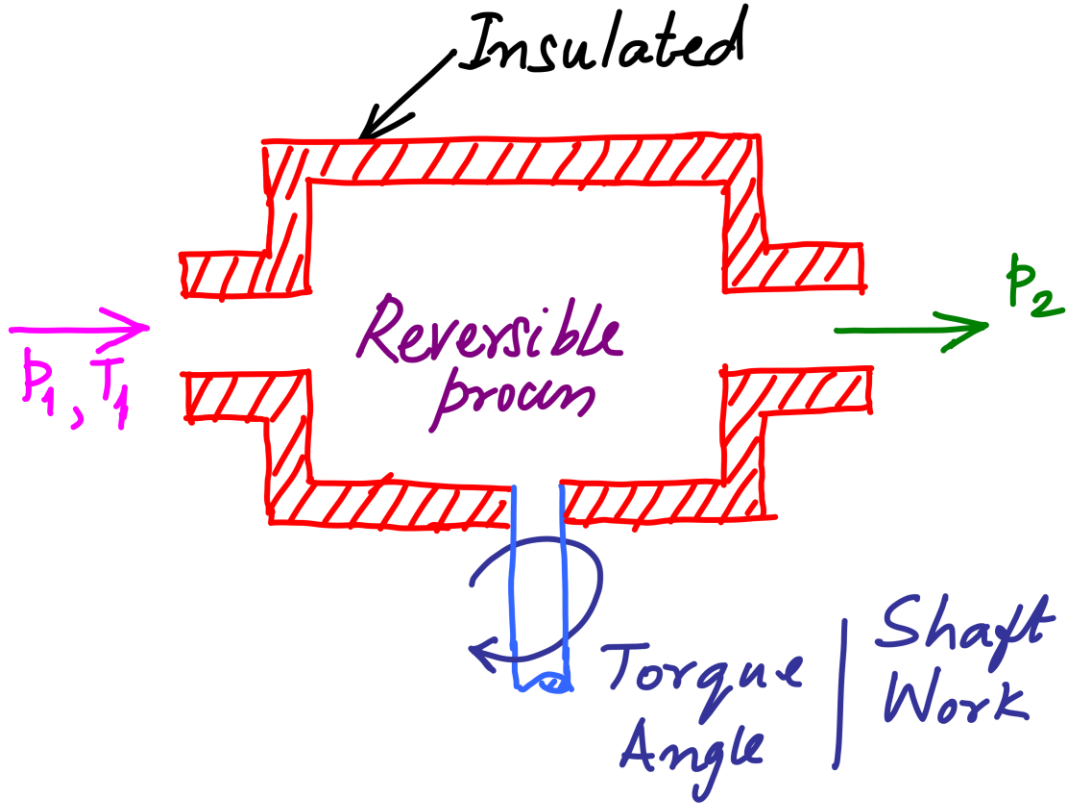


The outlet nozzles of the engines on a NASA F/A-18 Hornet
https://commons.wikimedia.org/w/index.php?title=File:Afterburner.jpg&oldid=559318095.

Isentropic efficiencies of Turbines, compressors and Nozzles

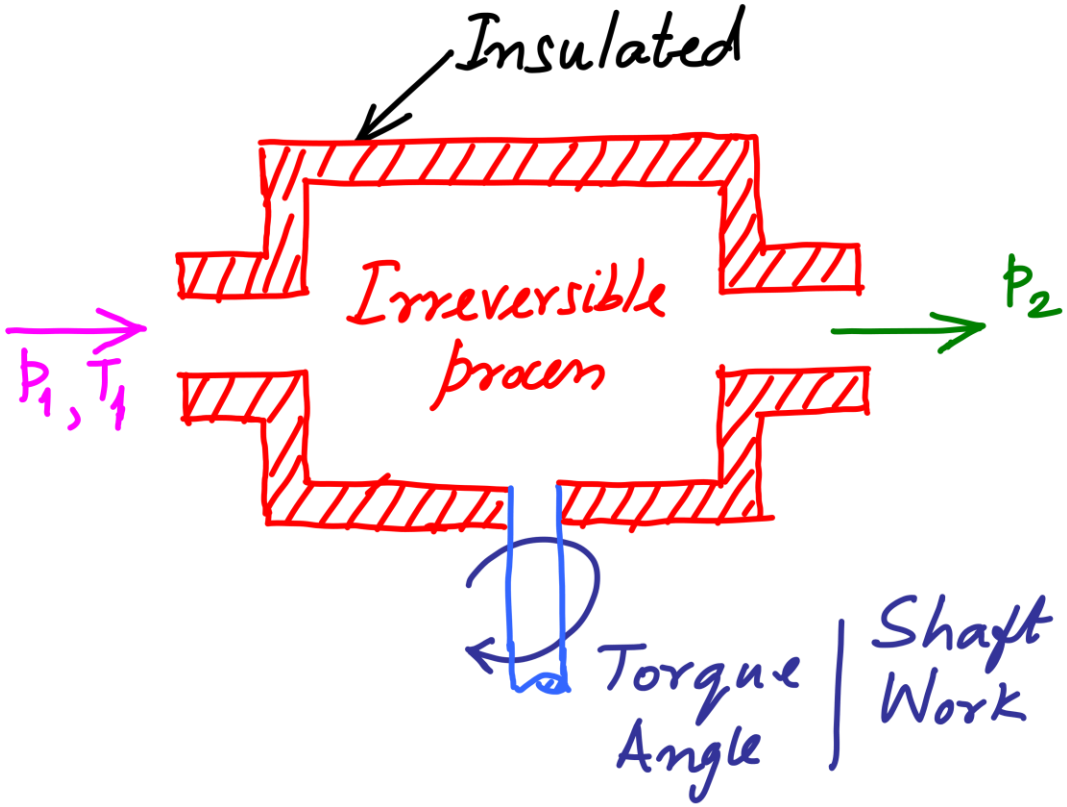
- Let us see how we can ascertain how well certain important steady flow devices perform
  - This will enable us to identify important processes that need attention in a large system that may be composed of these devices
  - Like an aircraft engine
- Hence we need to first define the limits of performance of these devices
  - If so we need to define the idealized processes that will best define the processes in these open systems or control volumes
- The limits are that these devices are supposed to operate without irreversibilities and also without heat transfer
  - Thus the Ideal process in these devices will be reversible adiabatic ***ISENTROPIC***

Figure of merit – Isentropic Efficiency of a turbine or compressor



Reference Process

Isentropic



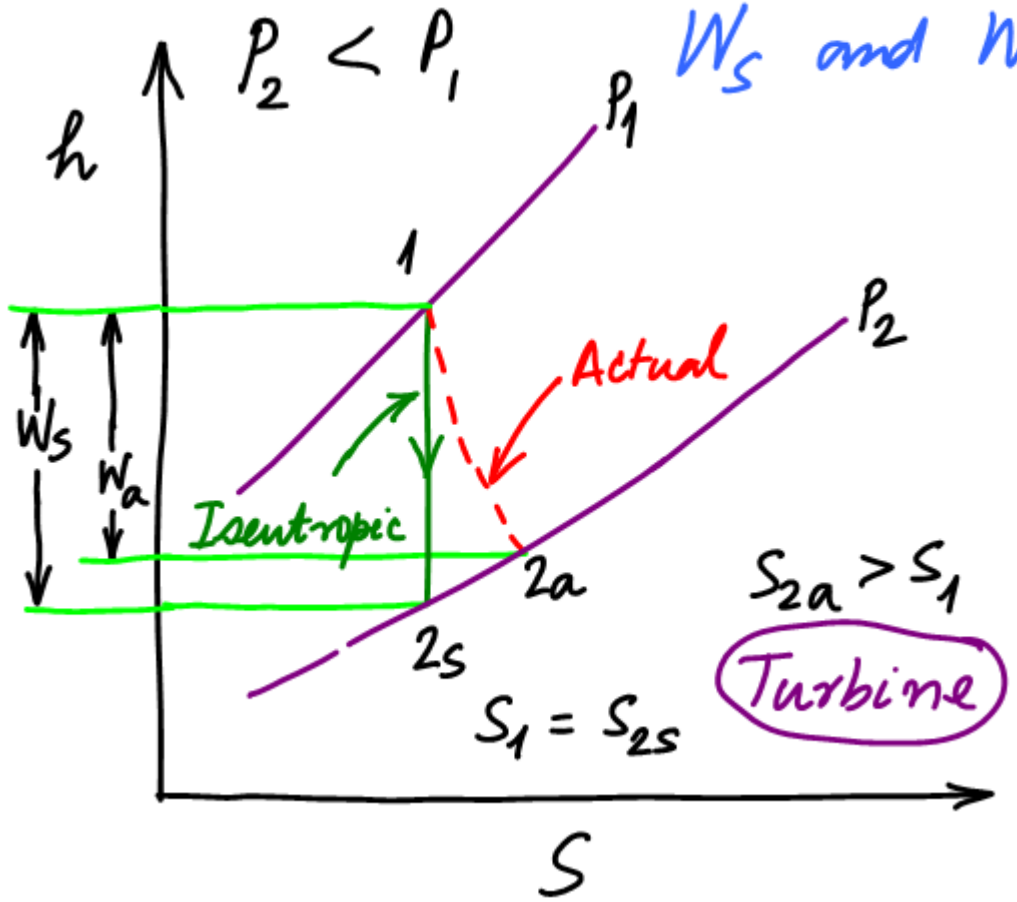
Actual Process

Adiabatic Irreversible

Turbine

- A steady flow device that produces shaft work outout at the expense of the enthalpy of the fluid that flows through it
- Generally considered to be adiabatic
- Kinetic energies at the inlet and exit generally considered to be negligible but have to be considered if otherwise
- Application of the first law under adiabatic steady flow conditions yields:
- $\frac{dm}{dt}\left[h_1 + \left(\frac{V_1}{2}\right)^2 + gz_1\right] + \Sigma\frac{\delta Q}{dt} = \frac{dm}{dt}\left[h_2 + \left(\frac{V_2}{2}\right)^2 + gz_2\right] + \frac{d(Wout_{turbine})}{dt}$
- Which is simplified as: $\frac{d(Wout_{turbine})}{dt} = \frac{dm}{dt}[h_1 - h_2]$

Isentropic Efficiency of a Turbine



W_s and W_a are Work outputs in a Turbine

$$W_s = h_1 - h_{2s}$$

$$W_a = h_1 - h_{2a}$$

$$\eta_{isen,turb} = \frac{W_a}{W_s} = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$$

For an ideal Gas with Constant Specific heats

$$= \frac{C_p(T_1 - T_{2a})}{C_p(T_1 - T_{2s})}$$

Process 1 – 2s

$$PV^\gamma = \text{Constant}$$

$$\left(\frac{T_{2s}}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\left(\frac{T_{2s}}{T_1}\right) = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

$\gamma = C_p/C_v$

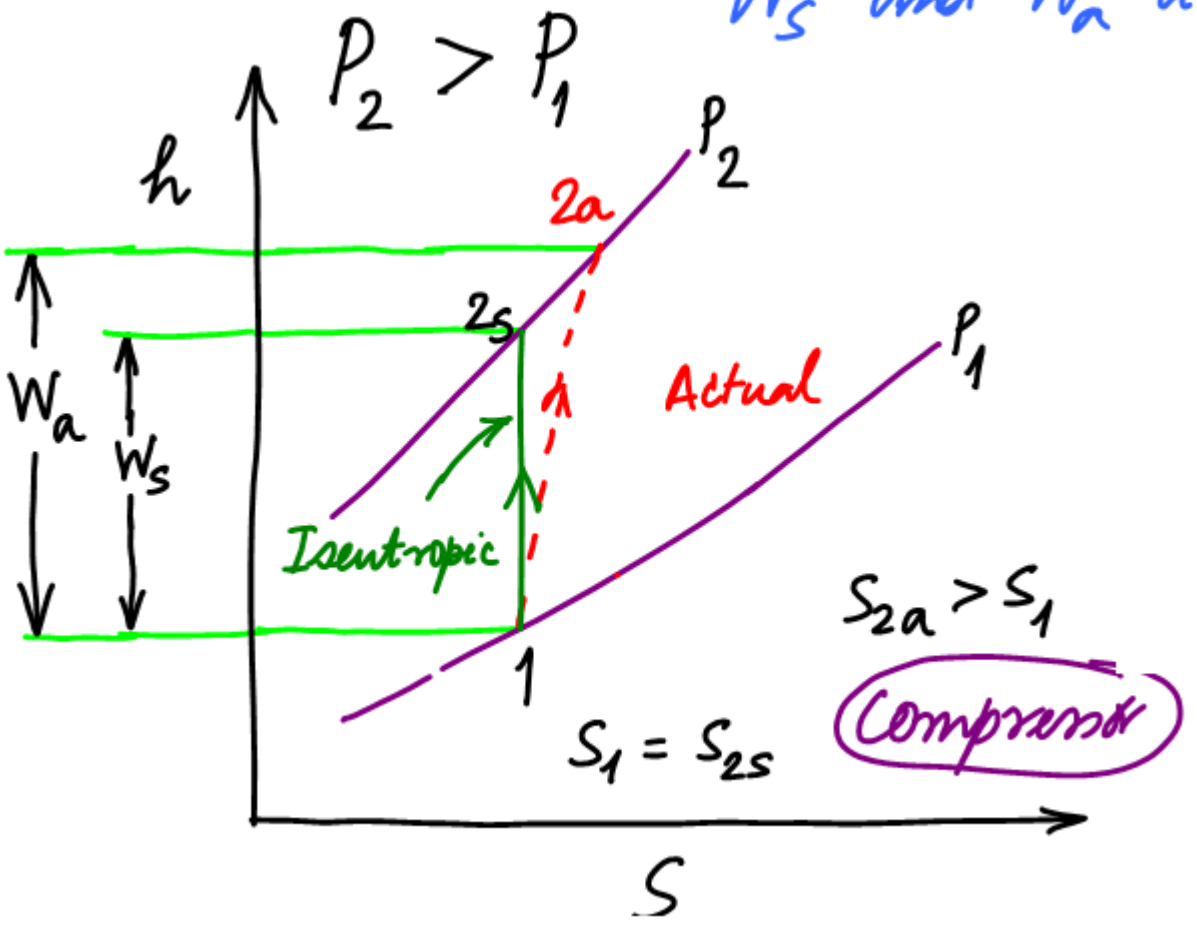
- **Adiabatic**
- **kinetic energies and elevation differences negligible**

Compressor

- A steady flow device that needs shaft work input and increases the pressure of the flowing fluid and hence its enthalpy
- Flow occurs from a lower to a higher pressure due to the work input
- Generally considered to be adiabatic
- Kinetic energies at the inlet and exit generally considered to be negligible but have to be considered if otherwise
- Application of the first law under adiabatic steady flow conditions yields:
- $\frac{dm}{dt}\left[h_1 + \left(\frac{V_1}{2}\right)^2 + gz_1\right] + \Sigma\frac{\delta Q}{dt} + \frac{dW_{in\ compressor}}{dt} = \frac{dm}{dt}\left[h_2 + \left(\frac{V_2}{2}\right)^2 + gz_2\right]$
- Which is simplified as: $\frac{dW_{in\ compressor}}{dt}$(taken to be positive)$=\frac{dm}{dt}\left[h_2 - h_1\right]$

Isentropic Efficiency of a Compressor

W_s and W_a are Work inputs in a Compressor



$$W_s = h_{2s} - h_1$$

$$W_a = h_{2a} - h_1$$

$$\eta_{isen,comp} = \frac{W_s}{W_a} = \frac{h_{2s} - h_1}{h_{2a} - h_1}$$

For an ideal Gas with Constant Specific heats

$$= \frac{C_p(T_{2s} - T_1)}{C_p(T_{2a} - T_1)}$$

Process 1 – 2s

$$PV^{\gamma} = \text{Constant}$$

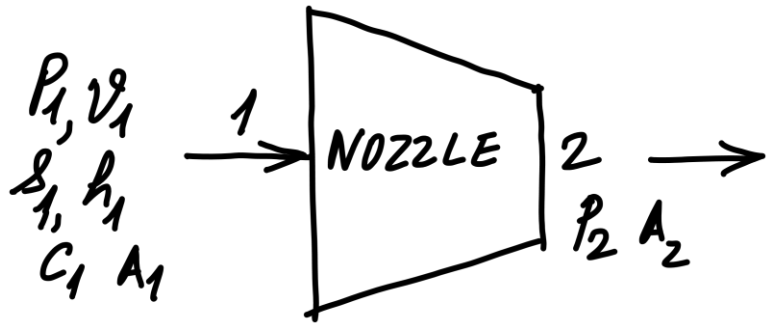
$$\left(\frac{T_{2s}}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\left(\frac{T_{2s}}{T_1}\right) = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

$\gamma = C_p/C_v$

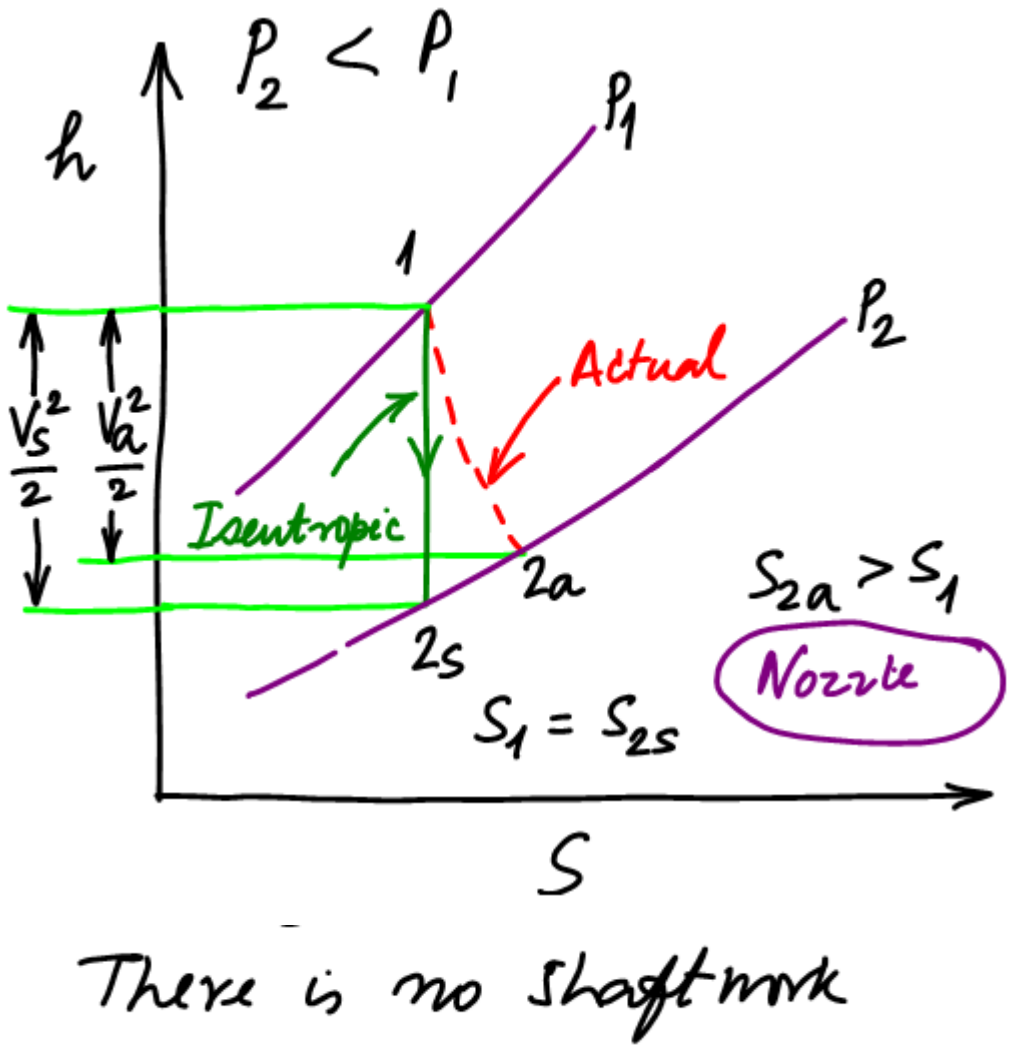
- **Adiabatic**
- **kinetic energies and elevation differences negligible**

Nozzle



- A steady flow device that is associated with no shaft work
- The pressure of the fluid that flows through it reduces from inlet to exit and its enthalpy also decreases.
- Flow occurs from a higher to a lower pressure and this increases the velocity of the fluid at the exit significantly as compared to the inlet
- Generally considered to be adiabatic
- Kinetic energy at the inlet generally considered to be negligible but has to be considered if otherwise. Kinetic energy at the exit is the real output
- Application of the first law under adiabatic steady flow conditions yields:
- $\frac{dm}{dt}\left[h_1 + \left(\frac{V_1}{2}\right)^2 + gz_1\right] + \Sigma\frac{\delta Q}{dt} = \frac{dm}{dt}\left[h_2 + \left(\frac{V_2}{2}\right)^2 + gz_2\right]$
- Which is simplified as: $\left(\frac{V_2}{2}\right)^2 = [h_1 - h_2]$, assuming $V_2 >> V_1$

Isentropic Efficiency of a Nozzle



There is no shaft work

$$\frac{V_s^2}{2} = h_1 - h_{2s}$$

$$\frac{V_a^2}{2} = h_1 - h_{2a}$$

$$\eta_{isen,noz} = \frac{(V_a^2/2)}{(V_s^2/2)} = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$$

For an ideal gas with constant specific heats

$$= \frac{C_p(T_1 - T_{2a})}{C_p(T_1 - T_{2s})}$$

Process 1–2s

$$PV^\gamma = \text{Constant}$$

$$\left(\frac{T_{2s}}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\left(\frac{T_{2s}}{T_1}\right) = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

$\gamma = C_p/C_v$

- **Adiabatic,**
- **inlet kinetic energy and elevation differences negligible**

Thank You